

GUIDANCE FOR ESTIMATING THE NUMBER OF CHILDREN IN NEED OF TREATMENT FOR WASTING



GUIDANCE DEVELOPMENT TEAM AND ACKNOWLEDGEMENTS

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Guidance for Estimating the Number of Children in Need of Treatment for Wasting

This document provides guidance on how to estimate the number of children with wasting and the expected total number of admissions for treatment.¹ Throughout this guidance, the term 'wasting' includes all forms of acute malnutrition measured by weight-for-height (WHZ), mid-upper arm circumference (MUAC) and/or bilateral pitting oedema.

The process of generating estimates of the number of children in need of treatment for wasting should be a collaborative process coordinated by the nutrition cluster/sector and involving key stakeholders (i.e., government, UNICEF, other United Nations agencies, NGOs and partners) responsible for delivering services for managing wasting. There should be consensus on the final estimates among the stakeholders. Individual agencies can also use this guidance. The primary purpose of generating estimates is to facilitate the planning, implementation, monitoring and evaluation of wasting treatment programmes.

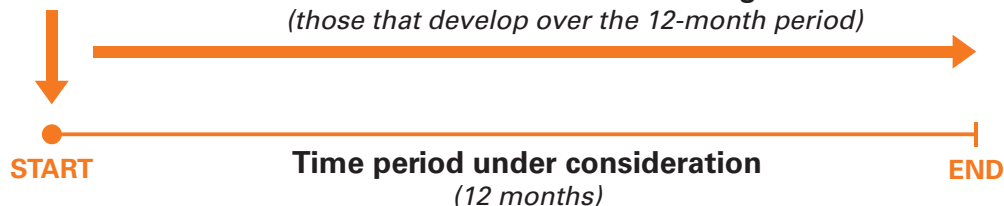
Background

The term 'wasting' captures both moderate and severe conditions. Children aged 6–59 months with a WHZ ≥ -3 and < -2 and/or a MUAC ≥ 115 mm and < 125 mm are classified as moderately wasted. Those with a WHZ < -3 and/or a MUAC < 115 mm and/or bilateral pitting oedema are classified as severely wasted.

For a new programme, estimates of the number of children with wasting over a specific time period should include prevalent cases (i.e., those that already exist at a specific time, typically determined using a representative household survey) and incident cases (i.e., new cases that develop during a specified time period, which is usually 12 months for wasting estimates). For estimates of the number of children with wasting over a specific time period within an existing programme, both the prevalent cases identified at the start of a specific period and the incident cases that will develop over the 12 month period need to be counted.

Prevalent cases of wasting

(those that exist at start of the 12-month period)



¹ Note that these were previously termed 'burden' and 'caseload', which are no longer used. In their place, use 'the number of children in need' of treatment and the 'expected total number of admissions' for treatment.

For advocacy, programme planning and implementation, it is important to estimate the **total number of children** with wasting that require treatment. Although the aspiration would be to reach all affected children; limited resources and access to affected populations means that programmes can reach only a proportion of these children. Therefore, two estimates are calculated:

1. The **total number of children aged 6–59 months with wasting in need of treatment** over a specific time period in a specific area (those that are suffering from wasting and meet the programme admission criteria). Referred to hereafter as **‘children in need’**.
2. The **total number of expected admissions of children aged 6–59 months with wasting** within a specific time period in a specific area (those with wasting that the programme expects to treat). Referred to hereafter as **‘expected admissions’**.²

The total number of expected admissions is projected from the total number of children in need using a treatment coverage estimate. Coverage surveys identify the proportion of children with wasting that are actually receiving treatment.

² This figure will support the calculation of the ‘number of children in need targeted’ (for UNICEF Humanitarian Action for Children (HAC) reports and Humanitarian Response Plans)

Calculating the number of children in need and the number of expected admissions

The recommended way to estimate the number of children in need and the number of expected admissions is to use programme admissions data from the previous 12-month period. Previous programme admissions data account for both incident cases (including a child admitted more than once in a 12-month period, counting as two admissions) and the programme’s treatment coverage. In the absence of data on previous admissions, such as in the case of new programmes or existing programmes with poor quality data, formulae are available to approximate estimates.

The Methods section below outlines the two recommended approaches in detail. The method used will depend on the availability and quality of programme data. For the purposes of this guidance, a 12-month planning period is used as the reference time period. Estimates are often required for a 12-month period, but the time period can be lengthened or shortened to suit programme planning needs.

Note that certain events or shocks (e.g., disease outbreaks or food insecurity) will affect wasting incidence and therefore estimates. Programmes will need to consider the potential impact of these events on the number of children in need and the number of expected admissions.

In this guidance, estimates refer to the number of children with wasting in need of treatment. However, over a 12-month period, it is possible that one child has two separate episodes of wasting due to the acute nature of the disease. Therefore, estimates of the ‘total number of children in need of treatment’ can also be interpreted as the ‘total number of cases in need of treatment’.

How to use this guidance document

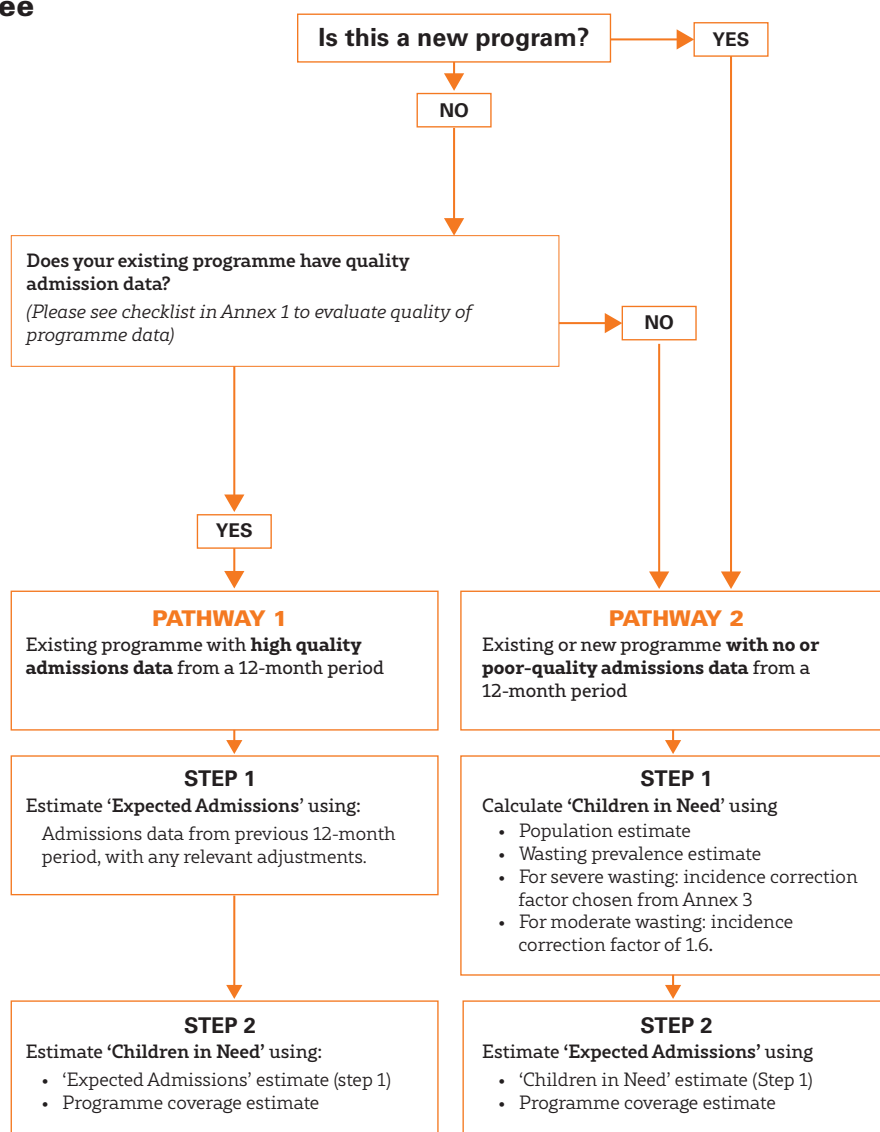
1 DECIDE: Review your data quality and availability. Use the decision tree below to identify the correct pathway to calculate the number of children in need and the number of expected admissions, based on data availability and quality. This will need to be done for severe wasting and moderate wasting separately.

2 DATA SOURCES: Ensure your data meet the requirements for each of the data components in your chosen pathway (see *Table 1*).

3 ESTIMATE: Calculate estimates for severe wasting and/or moderate wasting using the appropriate pathway.

4 REVIEW: Review and implement the guiding principles to improve data quality as required.

Decision Tree



Note: Due to the methods used, Pathway 1 calculates 'expected admissions' first, and Pathway 2 calculates 'Children in Need' first.

Methodological Note: Calculating a national figure using different pathways

In some countries, there may be both existing and new programme areas. In such cases, the sum of subnational estimates can be used to obtain a national figure. This is possible even if subnational estimates were calculated using different pathways. The approach used for aggregation at the national level should be clearly described in programme planning and reporting documents.

Pathway 1 should be used where possible, as it is considered the more feasible way to estimate the number of children in need of treatment and the number of expected admissions where there is an existing programme. For new programme areas, pathway 2 will have to be used for the first year until good quality programme admissions data covering a 12-month period become available; thereafter, pathway 1 should be used.

Data components

Table 1: Data Requirements for Estimate based by Pathway Chosen. Ideally all data points should be applicable to the same time period and geographic location

Required for pathway	Indicator	Definition	Source	Data requirements	Data management
1 only	Number of admissions to treatment programme for severe/moderate wasting	The number of severe/ moderate wasting cases aged 6–59 months of age admitted for treatment in a previous 12-month period in the specified geographic area	Routine programme data	- For severe wasting, includes both inpatient and outpatient admissions. - The case definition used should match the case definition that will be used for the next 12 months. - Includes all treatment providers in the geographic area.	- Information on the completeness of reporting (e.g., percent of facilities reporting) are required to understand the extent of missing data in the geographic catchment area. - If re-admission data are available, they should be included. If not, the data can still be used but this should be made clear in programme planning and reporting documents.
1 and 2	Percentage of severe/moderate wasting cases admitted to treatment programme (i.e., treatment point coverage estimate ¹)	Numerator: The number of children with severe/moderate wasting aged 6–59 month admitted to a treatment programme in a specific geographic catchment area at a single point in time Denominator: Total number of children with severe/moderate wasting in that geographical area at the same single point in time	Semi-Quantitative Evaluation of Access and Coverage (SQUEAC) or the Simple Spatial Survey Method (S3M) approaches are appropriate	- Point coverage with 95 percent confidence interval. - The survey should have been done within the geographic area and as near to the 12-month period covered by the programme admissions data as possible or in the last two years (when there have been no major changes to the context). - If multiple surveys are available for the same geographic area and time period, use the most recent survey. - If no coverage estimates are available for the time and place, use an average for all coverage surveys from the region or country. - SMART surveys are not currently recommended to provide point estimate for wasting coverage. Please see Annex 2 for guidance on how to use SMART surveys as a last resort if no coverage estimates are available.	For SQUEAC surveys without point coverage, calculate coverage using reported values of alpha prior, beta prior, number of severe/moderate wasting cases in treatment, and number of severe/moderate wasting cases out of treatment Do this using the Bayes SQUEAC calculator. See SQUEAC coverage survey guidance for more details.

Required for pathway	Indicator	Definition	Source	Data requirements	Data management
1 (optional for adjustment) and 2	Number of children aged 6–59 months	The estimated number of children aged 6–59 months living in the geographic area in the specified time period	Population estimates used in geographic area under consideration (e.g., adjusted Census estimates or health facility data)	- Represents the same geographic area and year as the prevalence survey. - When not available, use population data from the United Nations Population Division World Population Prospects.¹ In an emergency setting, use OCHA estimates if available.	
1 (optional for adjustment) and 2	Point prevalence of severe/moderate wasting in children aged 6–59 months	Numerator: The number of children aged 6–59 months with severe/moderate wasting in the geographic area Denominator: Total number of children aged 6–59 months in the geographic area	Representative household surveys, such as SMART (Standardized Monitoring and Assessment of Relief and Transition), Multiple Indicator Cluster Survey (MICS), Demographic and Health Survey (DHS) etc.)	- Children aged 6–59 months in a well-defined geographic area. - Survey is not older than two years old. - The estimates for children aged 6–59 months for a given geographical area should be representative of that population and the case definition for severe/moderate wasting should be specified. - Severe wasting in children aged 6–59 months can be defined as WHZ <-3 and/or MUAC <115 mm and/or bilateral pitting oedema. - Moderate wasting in children aged 6–59 months can be defined as WHZ ≥-3 and <-2 and/or MUAC ≥115 mm and <125 mm. - The survey case definition should align with the country-specific (or setting-specific) admission criteria. For example, if admitting only on MUAC, then use MUAC-only criteria to calculate point prevalence.	- For national estimates, the case definition for all prevalence estimates across provinces should be the same. - If there are two SMART surveys undertaken in one year for the same geographic area, review the contributing factors and drivers and use the point prevalence that is most representative of the current situation. This assumes there have been no significant changes in the context since the time of the selected SMART survey.
2	K-value	The K-value denotes the incidence correction factor. This factor, when combined with a prevalence estimate, enables an approximation of incident cases in a given period of time.	For severe wasting: K-value catalogue in Annex 3. For moderate wasting: K-value of 1.6	- For severe wasting, value ranges between 1 and 14.6 - For moderate wasting, value is 1.6 only.	If using pathway 2, choose K-value according to the guidance below.

¹**Point coverage** uses data for current cases only. It provides a snapshot of service performance at one point in time. **Period coverage** uses data for both current and recovering cases, providing estimates over a defined time period. See [here](#) for more information. Geographic coverage describes the proportion of service delivery units providing wasting services within the geographic area under consideration.

1 [United Nations Population Division World Population Prospects.](#)

METHODS

Methodological note: Inpatient and outpatient care

Estimates of the 'Number of expected admissions for severe wasting treatment' need to be disaggregated into inpatient and outpatient care.

If using pathway 1, estimates can be disaggregated according to the proportions observed in previous years. If using pathway 2, an estimated 85 percent of expected admissions will be for outpatient and 15 percent for inpatient care.

Severe wasting

Pathway 1: High quality programme admissions data are available

Step 1: Number of expected admissions for severe wasting treatment

Data required:

- Severe wasting programme admissions from the last 12-month period.
- If available, recent survey estimates that document any changes in severe wasting prevalence, severe wasting treatment programme coverage or population estimates since the last 12-month period. When there is no expected change of severe wasting prevalence, severe wasting treatment programme coverage or population estimates, this information is not needed.

For existing programmes with available data, the number of expected admissions (T_1) is likely to follow the number of admissions in the previous year (T_0), if there have been no contextual or epidemiological changes, so $T_1 = T_0$.

If there have been any recorded changes in the context since the last 12 months, these need to be accounted for in projections for the following 12 months. For example, there may be observed changes in prevalence, coverage or population estimates as indicated from recent surveys. The proposed adjustment will ensure that the admissions from the previous 12 months are adjusted to account for any new developments in the context, which could result in an increase or decrease in the number of expected admissions from the previous year. Changes in three factors, which have been used as examples, are accounted for in **Equation A**. This equation assumes that the correction factor K between prevalence and incidence is the same during the two periods.

Equation A

number of expected admissions (T_1)

$$= \text{Admissions}(T_0) \times \frac{\text{Population}^1}{\text{Population}^0} \times \frac{\text{Prevalence}^1}{\text{Prevalence}^0} \times \frac{\text{Coverage}^1}{\text{Coverage}^0}$$

Example

Data points required	Data from last year (0)	Data for next year (1)
Severe wasting treatment admissions (T)	4,073 children	Unknown
Population (6–59 months)	380,700 children	380,700 children †
Severe wasting prevalence	2.7%	4.1%*
Point coverage	34.9%	34.9%

$$\text{Number of expected admissions } (T_1) = 4073 \times \frac{380700}{380700} \times \frac{0.041}{0.027} \times \frac{0.349}{0.349}$$

$$\text{Number of expected admissions } (T_1) = 4073 \times 1 \times 1.52 \times 1$$

$$\text{Number of expected admissions } (T_1) = 6191$$

Interpretation: During the 12-month period under consideration, there will be an estimated 6191 admissions to either inpatient and outpatient treatment for severe wasting.

† Adjust as appropriate to consider any national population increases as per United Nations population estimates

*4.1% is the updated severe wasting prevalence

Step 2: Number of children in need of severe wasting treatment

Data required

- Estimate of number of expected admissions for severe wasting treatment (from step 1, above)
- Severe wasting treatment programme point coverage estimate

The number of children in need is directly related to the number of expected admissions by the inverse of point coverage. We need to inflate the estimate of the number of expected admissions (to 100 per cent programme coverage) to calculate the total number of children in need, using **Equation B**.

Frequently Asked Question: *If there is no coverage estimate for the programme area under consideration, can I still calculate the number of children in need of treatment?*

Yes, it is possible to use an average of all coverage surveys in the country; however, the decision to do so will depend on the context. Alternatively, it may be better to consider only recently collected data from neighbouring areas or areas with a similar context.

If the surveys use a range of methodologies (e.g., SQUEAC and SLEAC) and assess different administrative levels (e.g., district and region), all of the surveys can still be included in the average. If an average is used, this should be acknowledged in programme planning and reporting documents, as such estimates are likely to be less accurate than if a local coverage estimate was used. Assessing local programme coverage should be prioritized in the following year.

Equation B

$$\text{number of children in need} = \text{number of expected admissions} \times \frac{1}{\text{coverage}}$$

Example

Factors	Estimates
Number of expected admissions to severe wasting treatment (from step 1)	6191
Point coverage for severe wasting programme	34.9%

$$\text{Number of children in need} = 6191 \times \frac{1}{0.349}$$

$$\text{Number of children in need} = 17,739$$

Interpretation: During the 12-month period under consideration, the projected total number of children suffering from severe wasting in need of treatment within the programme area is 17,739.

Pathway 2: programme admissions data are not available or are of poor quality**Step 1 of Pathway 2: Number of children in need of severe wasting treatment**

For new programmes or existing programmes where high quality admissions data are not available, calculate the number of children in need using Equation c. This approach aims to account for both prevalent and incident cases using only readily available survey and census data an 'incidence correction factor' (denoted as 'K'). Prevalent cases can be obtained from prevalence survey estimates, and incident cases are approximated over a given period of time using the prevalence estimate and the incidence correction factor.

Data required

- Prevalence estimate of severe wasting among 6–59-month-olds (as detailed in Table 1).
- K-value chosen from Annex 3. It is not advised to calculate a K-value.

How to choose a K-value from Annex 3

1. Identify a K-value that reflects the context of the geographic area of interest in the time period under consideration. Prioritize choosing one from the same (i) province or (ii) country. Review the corresponding metadata for the K-value (including wasting prevalence, coverage estimate, presence of moderate wasting programme, IPC phase³) to determine the context of the value and therefore its applicability. Refer to in-country experts to determine the contextual differences (e.g., season, food security context, presence of acute emergencies). Multiple values may be available for the same geographic area in different time periods, in which case, choose a time period that closely reflects the current context.
2. In the absence of appropriate country-specific estimates, use (i) the regional K-value if it can be considered representative of the context, or (ii) the overall K-value. The selected K value used, and the justification should be documented in the planning and reporting documents.

Use **Equation C** to calculate the number of children in need over a 12-month period.

Frequently Asked Question: *If the source of wasting prevalence data varies for programme areas/provinces (e.g., SMART and MICS), can a mixture of data sources be used?*

The anthropometric indicator (WHZ, MUAC, oedema) used to estimate prevalence in the survey will impact the level of severe wasting observed. For example, if WHZ and MUAC are both assessed, the prevalence is likely to be higher than if WHZ only is used. The indicator used to calculate prevalence should reflect the programme admission criteria.

MICS and DHS surveys generally do not measure MUAC. If the wasting programme uses MUAC as an admission criteria, this mismatch will affect the calculation of the estimates.

A mixture of prevalence data survey methodologies can be used; however, they may not be directly comparable due to variable case definitions, which will affect the accuracy of the calculation. This should be made clear in programme planning and reporting documents.

Equation C

$$\text{Number of children in need} = \text{population}_{(6-59 \text{ months})} \times [\text{prevalence of severe wasting} \times (1 + K)]$$

³ Integrated Food Security Phase Classification

Example

Data points required	Data
Population 6-59 months	380,700
Prevalence severe wasting	2.7%
K-value (from Annex 3)	8

Number of children in need = $380,700 \times [0.027 \times (1 + 8)]$

Number of children in need = 92,510

Interpretation: During the 12-month period under consideration, there will be a total of 92,510 children suffering from severe wasting in need of treatment within the programme area.

Step 2: Number of expected admissions for severe wasting treatment**Data required**

- Number of children in need of severe wasting treatment (from step 1, above)
- Severe wasting treatment programme point coverage estimate

Once you have the estimate of the number of children in need, you can estimate the number of expected admissions by accounting for programme coverage using **Equation D**.

Equation D

Number of expected admissions = *number of children in need* × *point coverage*

For new programmes only, where treatment coverage has not yet been assessed through standard methodology, it is possible to approximate programme coverage using targets that are appropriate to the context. In the absence of coverage data, SPHERE⁴ standards can be used to estimate coverage using a coverage estimate of 50 per cent for rural settings, 70 per cent for urban settings and 90 per cent for camps.

Frequently Asked Question: *If there is no coverage estimate for the programme area under consideration, can I still calculate the number of children in need of treatment?*

Yes, it is possible to use an average of all coverage surveys in the country. If the surveys use a range of methodologies (e.g., SQUEAC and SLEAC) or assess different administrative levels (e.g., district and region), all of the surveys can be included in the average. If an average is used, acknowledge in programme planning and reporting documents as estimates are likely to be less accurate than a local coverage estimate. Assessing local programme coverage should be prioritized in the following year.

Example

Data points required	Data
Number of expected admissions to severe wasting treatment (from step 1)	92,510
Point coverage for severe wasting programme	34.9%

Number of expected admissions = $92,510 \times 0.349$

Number of expected admissions = 32,286

Interpretation: During the 12-month period under consideration, there will be an estimated 32,286 admissions to inpatient and outpatient treatment for severe wasting.

⁴ The SPHERE community sets standards for humanitarian action and promotes quality and accountability. See <<https://spherestandards.org/>>

Moderate wasting

The approach to estimate the MAM estimates for children in need and expected admission is similar to how SAM estimates were calculated, except for the choice of K.

Pathway 1: High quality programme admissions data are available

Step 1: Number of expected admissions for moderate wasting treatment

Data points required

- Moderate wasting programme admissions from the last 12-month period
- If available, recent survey estimates that document any changes in moderate wasting prevalence, moderate wasting treatment programme coverage or population estimates since the last 12-month period

For existing programmes with available data, the number of expected admissions (T_1) is likely to follow the number of admissions in the previous year (T_0) if there have been no contextual or epidemiological changes, so $T_1 = T_0$.

If there have been any recorded changes in the context during the last 12 months, these need to be accounted for in projections for the following 12 months. For example, there may be observed changes in prevalence, coverage or population estimates as indicated from recent surveys. The proposed adjustment will ensure that the admissions from the previous 12 months are adjusted to account for any new developments in the context, which could result in an increase or decrease in the number of expected admissions. Changes in three factors (population, prevalence and coverage), which have been used as examples, are accounted for in **Equation A**.

Frequently Asked Question: If there is no coverage estimate for the moderate wasting treatment programme, is it possible to use the coverage estimate for the severe wasting treatment programme?

No, we cannot assume the two programmes have the same level of coverage. Future coverage assessments should consider both the severe and moderate wasting treatment programmes.

Equation A

Number of expected admissions (T_1)

$$= \text{Admissions } (T_0) \times \frac{\text{Population}^1}{\text{Population}^0} \times \frac{\text{Prevalence}^1}{\text{Prevalence}^0} \times \frac{\text{Coverage}^1}{\text{Coverage}^0}$$

Example

Data points required	Data from last year (0)	Data for next year (1)
Moderate wasting treatment admissions (T)	7065 children	Unknown
Population (6-59 months)	380,700 children	380,700 children †
Moderate wasting prevalence	7.2%	10.0%*
Point coverage	40.1%	40.1%

$$\text{Number of expected admissions } (T_1) = 7065 \times \frac{380700}{380700} \times \frac{0.100}{0.072} \times \frac{0.401}{0.401}$$

$$\text{Number of expected admissions } (T_1) = 7065 \times 1 \times 1.38 \times 1$$

$$\text{Number of expected admissions } (T_1) = 9750$$

Interpretation: During the 12-month period under consideration, there will be an estimated 9750 admissions to outpatient treatment for moderate wasting.

† Adjust as appropriate to consider any national population increases as per United Nations population estimates

*10.0% is the updated moderate wasting prevalence

Step 2: Number of children in need of moderate wasting treatment**Data points required**

- Estimate of number of expected admissions for moderate wasting treatment (from step 1, above)
- Moderate wasting treatment programme point coverage estimate

The number of children in need is directly related to the number of expected admissions by the inverse of point coverage. We need to inflate the estimate of the number of expected admissions (to 100 per cent programme coverage) to calculate the total number of children in need, using **Equation B**.

Methodological Note

Using an incidence correction factor (K) of 1.6 for moderate wasting for all settings is sub-optimal. It is suggested for limited use here due to the current absence of moderate wasting-specific estimates of K. As soon as admissions data are available in a given context, pathway 1 can be used.

Equation B

$$\text{number of children in need} = \text{number of expected admissions} \times \frac{1}{\text{coverage}}$$

Example

Data points required	Estimates
Number of expected admissions for moderate wasting treatment (from step 1)	9750
Point coverage for moderate wasting programme	40.1%

$$\text{Number of children in need} = 9750 \times \frac{1}{0.401}$$

$$\text{Number of children in need} = 24,314$$

Interpretation: During the 12-month period under consideration, the projected total number of children suffering from moderate wasting in need of treatment within the programme areas is 24,314.

Pathway 2: Programme admissions data are not available or are of poor quality**Step 1: Number of children in need of moderate wasting treatment**

For new programmes or existing programmes where high quality admissions data are not available, calculate the number of children in need using **Equation C**. This approach aims to account for both prevalent and incident cases using only readily available survey and census data and an 'incidence correction factor' (denoted as 'K'). Prevalent cases can be obtained from prevalence survey estimates, and incident cases are approximated over a given period of time using the prevalence estimate and the 'incidence correction factor'.

Data points required

- Population aged 6–59 months
- Prevalence estimate of moderate wasting (WHZ ≥ -3 and < -2 and/or MUAC ≥ 115 mm and < 125 mm)
- K-value of 1.6

Equation C

$$\text{Number of children in need} = \text{population}_{(6-59 \text{ months})} \times [\text{prevalence of moderate wasting} \times (1 + K)]$$

Or simplified to be:

$$\text{Number of children in need} = \text{population}_{(6-59 \text{ months})} \times [\text{prevalence of moderate wasting} \times (1 + 1.16)]$$

Example

Data points required	Estimates
Population 6–59 months	380,700
Prevalence of moderate wasting	8.1%
Incidence correction factor (K)	1.6

Number of children in need = 380,700 × (1 + 1.6)

Number of children in need = 80,176

Interpretation: During the 12-month period under consideration, there will be a total of 80,176 children suffering from moderate wasting in need of treatment within the programme area.

Step 2: Number of expected admissions for moderate wasting treatment

Data points required

- Estimate of number of children in need of moderate wasting treatment (from step 1, above)
- Moderate wasting treatment programme point coverage estimate

Once you have the estimate of the number of children in need, you can estimate the number of expected admissions by accounting for programme coverage using **Equation D**.

Equation D

Number of expected admissions = *number of children in need* × *point coverage*

For new programmes only, where treatment coverage has not yet been assessed through standard methodology, it is possible to approximate programme coverage using targets that are appropriate to your context. SPHERE standards can be used to estimate coverage using 50 per cent for rural settings, 70 per cent for urban settings and 90 per cent for camps.

Example

Data points required	Estimates
Number of children in need (from step 1)	80,176
Point coverage for moderate wasting programme	40.1%

Number of expected admissions = 80,176 × 0.401

Number of expected admissions = 32,150

Interpretation: During the 12-month period under consideration, there will be an estimated 32,150 admissions to outpatient treatment for moderate wasting.

Methodological note: Calculating estimates for more or less than a 12-month period

- If estimates are required for a specific period while using pathway 1, simply identify admissions from the months of interest only, before starting any calculations. For example, if you want an estimate of expected admissions for treatment between January and June only, identify admissions from January to June in the previous year.
- While using pathway 2, identify the proportion of all months that are of interest and use that as the estimate. For example, if an estimate is required for 6 months, divide the 12-month estimate by two (e.g., 12 months / 6 months). It is important to note that this method does not adequately account for seasonal variation.

GUIDING PRINCIPLES FOR IMPROVING DATA QUALITY

To determine the estimates of the number of children in need and expected admissions for programme planning and evaluation – good quality data that is context specific is vital. This section provides broad guidance on improving routine data collection. Please note that this section will be updated to provide more detailed guidance on how to improve data quality.

While calculating estimates, partners should identify and document data related gaps and challenges with the aim to address them as soon as possible. Indicators to measure the quality and effectiveness of wasting treatment programmes are globally defined. Partners should work together as a priority to support and strengthen government-owned nutrition information systems to improve the availability and quality of nutrition programme data.

Routine programme data

1. UNICEF and partners are developing a standardized guide on core nutrition indicators to be included in routine information systems such as the DHIS2. This guide will provide the standardized templates for collection of nutrition programme data. The guidance and standardized templates will enable the collection of accurate programme data, countries are encouraged to use the standardized templates to collect data once available.
2. The following data should be included in routine monitoring systems, with the following rationale:
 - a. Screening data: Wasting programmes will screen children to ensure children that require treatment are included in the programmes. The screening can be done at health facility or community level and involves measuring children using MUAC or WHZ indicators to check for wasting. This information on the number of children screened, the number wasted and referred for treatment will inform managers on the reach of the programme and is useful for programme planning purposes.
 - b. Age disaggregated data: Children under 2 years of age are more vulnerable to malnutrition and at a higher risk of suffering its long-term and irreversible effects such as stunting and mortality. Therefore, it is essential to report data for children aged 6–23 months and 24–59 months separately. If services are provided for

infants under 6 months of age, report this data separately.

- c. Admission criteria: To ensure that a programme is capturing all expected admissions, the criteria on which the child was admitted (WHZ/MUAC/oedema) should be recorded in both registers and summary tools. Often these criteria are excluded from summary tools, which limits analysis of programme performance.
- d. Indicators on stock management: Information on stock management is critically important to ensure that health facilities have the required stocks to treat the total number of expected treatment admissions.

Prevalence and coverage surveys

3. Start systematically collecting data on infants under 6 months of age in SMART surveys. Suggested indicators to be collected for this age group are weight-for-age <-2 and MUAC. Although there are not yet internationally agreed cut-offs for MUAC for infants under 6 months of age, the suggested criteria for SAM are as follows (based on the best available evidence to date):
 - For infants < 6 weeks of age – use MUAC <110 mm
 - For infants ≥ 6 weeks of age – use MUAC <115 mm

The recommendation is to report the prevalence of SAM using these thresholds; however, thresholds may change as more evidence becomes available. Note that it is important to collect data on the age profile of the infants assessed in the survey to aid interpretation of the context. Further, as the data for this age group will be analysed separately from that of older children (those aged 6–59 months), it is important to ensure that the sample size of children < 6 months is sufficient for estimation.

4. Report moderate and severe wasting prevalence as the percentage of all children identified either by WHZ and/or MUAC and/or oedema, rather than using a single indicator (e.g., WHZ only).
5. Continue efforts to address the quality of anthropometric measurements in surveys through rigorous training of enumerators, strong active supervision and regular data quality checks during fieldwork. Accurate data are required for generation

- of reliable estimates and facilitate efficient programme management.
6. Budget for coverage surveys in monitoring and evaluation plans. Coverage surveys are designed to provide an estimate of programme coverage and identify the key barriers/bottlenecks affecting the programme as well as the factors that boost coverage. The aim of the coverage survey is to improve the coverage and quality of the programme.
 7. Consider aligning coverage surveys with the same geographic areas as prevalence surveys to the extent possible within financial and operational constraints. This will provide more valid estimates of the number of children in need and the number of expected admissions.
 8. The case criteria used for both surveys (prevalence and coverage) and programme admissions should be the same. This will ensure estimates of the number of children in need, expected admissions and children reached are aligned and therefore comparable.
 9. Establish triggers to determine when repeated or new prevalence and coverage surveys are necessary. Having regular and timely prevalence and coverage estimates are required to inform more accurate programme management and estimates.

ANNEX 1: ASSESSING QUALITY OF PROGRAMME DATA

Guiding points to assess quality of programme data to be used for estimating number of children in need of wasting treatment

This section will be further developed to provide more specific guidance to evaluate the quality of programme data. Currently there is limited guidance on methods and cut-offs/thresholds to determine the quality of nutrition programme data collected through routine systems. The points below provide overall considerations to guide the users to assess if the quality of the available programme data is reliable to be used to calculate the estimates.

The purpose of the checklist is to provide a general list of the key components to support the review of the availability and quality of wasting programme admission data for the estimation of the number of children with wasting in need of treatment using **Pathway 1**. The quality and availability of data for the area being assessed will influence the final estimation and therefore it is essential to have an understanding of the quality of the data.

The details of the data requirements and considerations are found in the table below:

a. Completeness of monthly reports

Data requirements		Key data considerations
Reporting rates	The number of health facilities that are reporting will impact the overall data available for calculating the estimates	What is the reporting rate for the health facilities in the area being assessed ? Review the percentage of health facilities reporting monthly for the area being assessed. Reporting rates of over 80% are generally satisfactory
Availability of data	Availability of data to ensure there is a full set of data for 12 months for the area being assessed	Is there wasting programme data available for the last 12 months? If data are not available for all areas being accessed, or if the records have data gaps, the estimates will be inaccurate. It will not be possible to calculate estimates without wasting programme data for 12 months.

These guiding points should be reviewed by the persons conducting the calculations. It is important to take note of the key issues arising in the quality of the data; this information should be acknowledged in programme planning and reporting documents and shared with the key stakeholders to address the data gaps as soon as possible.

The four key categories to guide users in reviewing availability and quality of available data are:

- a. Completeness of monthly reports
- b. Accuracy of monthly reports
- c. Adherence to programme and monitoring protocols
- d. Existence of quality assurance mechanisms in the data system

The availability of quality programme data will include data that have high reporting rates for areas being assessed, minimal data gaps, no duplicates, is timely (*from within the last year at least*) and is consistent with expected programme implementation. Reviewing the adherence to programme and monitoring protocols and the existence of a quality assurance mechanism will provide a deeper understanding of the programme data and quality.

b. Accuracy of monthly reports

Data requirements		Key data considerations
Trends over time	Analyse the data trends for the area under assessment to understand changes over time and their impact on the interpretation and quality of the data	Are the trends observed in the data due to seasonality or other reasons like programme related activities and reporting? Review data trends to check for data consistency over time. The changes in trends over time should be analysed and triangulated with other data sources to understand: • Seasonality patterns • Programme related activities • Reporting gaps
Data accuracy	Review the datasets for numbers that are unrealistic, missing, or inconsistent with programme expectations	How accurate are the numbers reported? Check the data sets for: Implausible numbers • Missing data – are data sets complete or are there key data gaps within the data sets • Duplication of data – check if there are duplicate entries • Outdated data – data should be at most one year old Understand if the data estimates are consistent with other programme activities (e.g., supplies used) Investigate data showing extremely high cure rates in areas with limited access

c. Adherence to programme and monitoring protocols

Use of national programme guidelines or recommendations	Review the national programme guidelines and their implementation	Is the programme adhering to the national guidelines for treatment, in terms of admission criteria, discharge etc.?
Availability of measuring equipment	Consider the availability of resources – equipment and capacity for anthropometric measurements in facilities	• Do most facilities have the anthropometric equipment required for taking measurements? • Are the enumerators trained regularly?
Reporting guidelines and tools	Review the availability of reporting guidelines and tools to support the monitoring and reporting processes	• Is the reporting process clear and well documented? • Do most facilities have the forms and reporting tools required for developing and submitting reports?

d. Existence of quality assurance mechanisms on the data system

Monitoring and quality assurance processes	Existence of data quality assurance processes for the systematic review of data and information collected through routine systems	• Is there a process for the systematic review of the data and reporting procedures? • Are there quality assurance processes done systematically at district/ subnational level? • How often are the quality assurance processes conducted?
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ANNEX 2: USE OF SMART SURVEYS TO ESTIMATE COVERAGE WHEN NO PROGRAMME COVERAGE ESTIMATES ARE AVAILABLE

SMART surveys are not currently recommended to provide point estimate for SAM or MAM coverage. This is because the small sample size/number of SAM cases (or MAM cases) detected from a SMART survey is likely make the coverage estimates imprecise. In addition, SMART surveys use the Probability Proportional to Size (PPS) sampling methodology, which may not accurately consider geographic variations in coverage in some settings. However, in situations where there is absolutely no other way to estimate programme coverage, SMART surveys can be used as a last resort.

SMART survey data could be used to as a proxy for program coverage provided that the **minimum** number of SAM or MAM cases is 40 each in the given SMART survey. Note that it is unlikely that the number of SAM cases would be 40 in a given SMART survey, since this would translate to a SAM prevalence of about 7% if the survey's sample size was 600. However, in the case that either SAM or MAM cases are 40 or more, then the coverage estimate can be calculated.

With a minimum number of 40 cases, the precision around the coverage estimate would be $\pm 15\%$ if the coverage is about 50% (if the coverage is different from 50%, then the precision will be higher). Although a precision of $\pm 15\%$ yields a large confidence interval, it is still useful for interpreting coverage estimate – for instance, if the coverage is estimated as 60% using the observed 40 cases, then there is reasonable certainty that the coverage is above 45%; and similarly, if the coverage is 35% using the observed 40 cases, then there is reasonable certainty that the coverage doesn't exceed 50%.

The table below provides the estimated precision which would be achieved from specific number of SAM or MAM Cases assuming a coverage estimate of 50%. Forty cases are required for $\pm 15\%$ precision around the coverage estimate, and while $\pm 10\%$ precision around the coverage estimate can be achieved using 90 cases.

Estimated precision given specific number of SAM or MAM Cases

Number of Cases	Precision
40 (Minimum)	15%
50	14%
60	13%
70	12%
80	11%
90	10%
100	10%

To estimate coverage from a SMART survey, an additional question in the questionnaire (data collection tool) is required. The question could be framed as below:

*Is this child currently registered in any nutrition programme?
(Confirmation should be done via presentation of a beneficiary card to ascertain if the child is in the program, and the type of program eg. SAM or MAM)*

Response to this additional question in the questionnaire (data collection tool) will allow coverage to be estimated with the formula below:

$$\text{Program Coverage}^* = \frac{\text{Number of SAM or MAM children currently in the program (either SAM or MAM Program)}}{\text{Total Number of Children classified as GAM (SAM+MAM)}}$$

*If SAM cases in the survey are 40 and above; then SAM Coverage can be estimated

ANNEX 3: CATALOGUE OF K-VALUES

Information on the study and the methodology used is available via: Isanaka, S. and Andersen, CT, *Improving estimates of the burden of wasting using incidence data*, Boston: Harvard T.H. Chan School of Public Health, 2020. Available at: [Isanaka S, Andersen CT, Cousens S, Myatt M, Briend A, Krasevec J, Hayashi C, Mayberry A, Mwirigi L, Guerrero S. Improving estimates of the burden of severe wasting: analysis of secondary prevalence and incidence data from 352 sites. BMJ Glob Health. 2021 Mar;6\(3\):e004342.](#)

* Average of all coverage surveys for the country. † SAM defined as MUAC < 110 mm, WHZ < -3 and/or oedema. Regional estimates include estimates from all countries listed in the region. To reduce the influence of implausible K estimates on summary values at the national or regional level, K estimates were excluded if too high (i.e., according to the Tukey statistical criteria, $K > 15.2$) or too low (i.e., according to expert consensus, $K < 1$).

The last four columns – ‘IPC phase’, ‘prevalence estimate taken during lean season?’, ‘MAM treatment programme present?’ and ‘acute emergency?’ – provide information about the context at the time that the data were collected, which was used to calculate the K-value. This information should be reviewed when considering whether the K-value is relevant to the current context, as these factors will all affect incidence. For example, is the IPC phase that the area is currently in the same as it was when the data were collected?

Location & date	Severe Wasting treatment admissions (n)	Coverage estimate (%)	Estimated incident cases (n)	Population (n)	Prevalence (%)	Prevalent cases (n)	K-value for severe wasting estimates	IPC phase at the time of data collection	Prevalence estimate taken during lean season? (Y/N)	Moderate Wasting treatment programme present?	Acute emergency (Y/N)
Overall	283	-	-	-	-	-	3.6 (3.4, 3.9)		-	-	-
West and Central Africa	147	-	-	-	-	-	3.8 (3.4, 4.2)		-	-	-
Burkina Faso	30	-	-	-	-	-	3.9 (3.4, 4.6)		-	-	-
Boucle du Mouhoun Region, Sep 2017	8,467	36.7 (26.2, 47.5)*	23,040	324,708	1.5 (1.1, 2.1)	4,967	4.6 (3.1, 7.4)	1	-	-	-
Cascades Region, Sep 2017	3,023	36.7 (26.2, 47.5)*	8,226	125,204	2.2 (1.4, 3.4)	2,760	3.0 (1.8, 5.3)	1	-	-	-
Centre Est Region, Sep 2017	3,955	36.7 (26.2, 47.5)*	10,762	269,167	1.8 (1.2, 2.8)	4,939	2.2 (1.4, 3.9)	1	-	-	-
Centre Nord Region, Sep 2017	9,455	36.7 (26.2, 47.5)*	25,728	287,866	1.7 (1.1, 2.7)	4,959	5.2 (3.0, 8.9)	1	-	-	-
Centre Ouest Region, Sep 2017	8,605	36.7 (26.2, 47.5)*	23,415	268,230	1.5 (0.8, 3.0)	4,041	5.8 (2.8, 13.4)	1	-	-	-
Centre Region, Sep 2017	4,525	36.7 (26.2, 47.5)*	12,313	336,474	3.2 (1.5, 6.6)	10,755	1.1 (0.5, 2.7)	1	-	-	-
Centre Sud Region, Sep 2017	1,664	36.7 (26.2, 47.5)*	4,528	140,340	1.8 (1.1, 2.8)	2,478	1.8 (1.1, 3.1)	1	-	-	-
Est Region, Sep 2017	11,650	43.5 (33.9, 53.1)	26,774	314,967	2.3 (1.7, 3.0)	7,184	3.7 (2.7, 5.4)	1	-	-	-
Hauts Bassins Region, Sep 2017	6,973	36.7 (26.2, 47.5)*	18,974	300,234	2.1 (1.3, 3.2)	6,155	3.1 (1.8, 5.5)	1	-	-	-
Nord Region, Sep 2017	11,838	36.7 (26.2, 47.5)*	32,213	276,892	1.7 (1.0, 2.8)	4,728	6.8 (3.8, 12.5)	1	-	-	-
Plateau Central Region, Sep 2017	3,598	36.7 (26.2, 47.5)*	9,791	158,430	1.3 (0.8, 2.2)	2,118	4.6 (2.6, 8.6)	1	-	-	-
Sahel Region, Sep 2017	12,230	36.7 (26.2, 47.5)*	33,279	228,938	4.5 (3.5, 5.8)	10,368	3.2 (2.2, 4.9)	1	-	-	-

ESTIMATING THE NUMBER OF CHILDREN IN NEED OF TREATMENT FOR WASTING

Location & date	Severe Wasting treatment admissions (n)	Coverage estimate (%)	Estimated incident cases (n)	Population (n)	Prevalence (%)	Prevalent cases (n)	K-value for severe wasting estimates	IPC phase at the time of data collection	Prevalence estimate taken during lean season? (Y/N)	Moderate Wasting treatment programme present?	Acute emergency (Y/N)
Sud-Ouest Region, Sep 2017	4,987	36.7 (26.2, 47.5)*	13,570	141,686	1.6 (0.7, 3.8)	2,249	6.0 (2.4, 15.2)	1	-	-	-
Boucle du Mouhoun Region, Sep 2016	9,058	36.7 (26.2, 47.5)*	24,648	319,113	2.1 (1.3, 3.2)	6,662	3.7 (2.2, 6.7)	1	-	-	-
Cascades Region, Sep 2016	3,367	36.7 (26.2, 47.5)*	9,162	122,073	1.9 (1.0, 3.5)	2,286	4.0 (2.1, 8.0)	1	-	-	-
Centre Est Region, Sep 2016	5,010	36.7 (26.2, 47.5)*	13,633	263,783	1.4 (0.7, 2.6)	3,581	3.8 (1.9, 8.4)	1	-	-	-
Centre Nord Region, Sep 2016	9,874	36.7 (26.2, 47.5)*	26,868	282,362	0.7 (0.3, 1.7)	2,064	13.0 (5.1, 31.5)	1	-	-	-
Centre Ouest Region, Sep 2016	10,066	36.7 (26.2, 47.5)*	27,391	263,377	1.4 (0.8, 2.6)	3,736	7.3 (3.5, 15.4)	1	-	-	-
Centre Region, Sep 2016	3,544	36.7 (26.2, 47.5)*	9,644	326,885	1.7 (0.9, 3.5)	5,697	1.7 (0.9, 3.5)	1	-	-	-
Est Region, Sep 2016	11,384	36.7 (26.2, 47.5)*	30,977	307,976	3.0 (2.0, 4.3)	9,120	3.4 (2.1, 5.5)	1	-	-	-
Hauts Bassins Region, Sep 2016	7,206	36.7 (26.2, 47.5)*	19,608	293,835	0.9 (0.4, 1.9)	2,526	7.8 (3.3, 17.6)	1	-	-	-
Nord Region, Sep 2016	13,197	36.7 (26.2, 47.5)*	35,911	271,931	2.1 (1.3, 3.3)	5,620	6.4 (3.7, 11.8)	1	-	-	-
Plateau Central Region, Sep 2016	4,203	36.7 (26.2, 47.5)*	11,437	155,695	1.3 (0.6, 2.6)	1,964	5.8 (2.6, 13.7)	1	-	-	-
Sahel Region, Sep 2016	13,356	36.7 (26.2, 47.5)*	36,343	224,193	2.1 (1.2, 3.6)	4,623	7.9 (4.2, 14.8)	1	-	-	-
Sud-Ouest Region, Sep 2016	4,878	36.7 (26.2, 47.5)*	13,274	139,045	2.7 (1.5, 5.0)	3,778	3.5 (1.8, 6.9)	1	-	-	-
Countrywide, Nov 2015	103,542	48.8 (38.1, 59.5)	212,327	3,044,336	2.3 (2.1, 2.6)	70,892	3.0 (2.4, 4.0)	1	-	-	-
Countrywide, Sep 2014	116,262	36.7 (26.2, 47.5)*	316,363	2,982,797	1.8 (1.5, 2.1)	54,082	5.8 (4.3, 8.4)	2	-	-	-
Countrywide, Sep 2013	104,582	40.9 (30.6, 51.3)	255,551	2,921,111	2.0 (1.6, 2.3)	57,125	4.5 (3.4, 6.4)	1	-	-	-
Countrywide, Sep 2012	91,891	37.8 (26.5, 49.0)	243,171	2,860,908	2.3 (2.1, 2.5)	65,543	3.7 (2.8, 5.4)	1	-	-	-
Countrywide, Sep 2011	54,613	28.9 (17.1, 41.7)	188,931	2,783,611	2.6 (2.3, 3.0)	73,652	2.6 (1.8, 4.5)	2	-	-	-
Chad	7						6.6 (3.2, 13.6)				
Région de Logone orientale, Aug 2016	7,340	37.0 (26.7, 48.1)*	19,827	172,563	2.1 (1.3, 3.5)	3,666	5.4 (3.0, 10.1)	1	Y	-	-
Région de Mandoul, Aug 2016	2,877	37.0 (26.7, 48.1)*	7,771	141,720	4.9 (3.5, 6.7)	6,874	1.1 (0.7, 1.8)	1	Y	-	-
Région de Batha, Jan 2014	12,621	19.7 (10.9, 28.5)	64,095	111,353	5.3 (3.5, 7.8)	5,861	10.9 (6.3, 21.6)	2	N	-	-
Région de Hadjer Lamis, Jan 2014	9,028	40.3 (30.3, 50.3)	22,405	118,409	2.7 (1.6, 4.5)	3,176	7.1 (3.9, 12.5)	1	N	-	-
Région de Lac, Jan 2014	12,651	37.0 (26.7, 48.1)*	34,173	95,353	2.4 (1.4, 3.9)	2,247	15.2 (8.3, 26.7)	1	N	-	-
Région de Salamat, Jan 2014	8,876	43.1 (32.6, 53.6)	20,610	64,759	3.7 (2.7, 5.1)	2,396	8.6 (5.6, 13.4)	1	N	-	-
Région de Sila, Jan 2014	8,948	40.9 (29.9, 51.9)	21,859	53,470	4.1 (2.8, 6.1)	2,216	9.9 (6.3, 16.8)	1	N	-	-

ESTIMATING THE NUMBER OF CHILDREN IN NEED OF TREATMENT FOR WASTING

Location & date	Severe Wasting treatment admissions (n)	Coverage estimate (%)	Estimated incident cases (n)	Population (n)	Prevalence (%)	Prevalent cases (n)	K-value for severe wasting estimates	IPC phase at the time of data collection	Prevalence estimate taken during lean season? (Y/N)	Moderate Wasting treatment programme present?	Acute emergency (Y/N)
Democratic Republic of Congo	3						4.9 (1.9, 12.6)				
Banyakiri health zone, Sep 2017	1,985	34.2 (23.9, 44.6)*	5,804	41,737	2.2 (1.4, 3.6)	929	6.2 (3.6, 11.6)	1	Y	-	-
Banyakiri health zone, Jan 2016	1,514	34.2 (23.9, 44.6)*	4,427	37,444	5.4 (4.1, 7.2)	2,029	2.2 (1.4, 3.4)	-	N	-	-
Kalomba health zone, Oct 2015	2,262	37.9 (28.2, 47.5)	5,973	25,434	2.3 (1.3, 4.0)	588	10.2 (5.4, 19.3)	-	Y	N	N
Ghana	3						1.9 (1.0, 3.9)				
Northern Region, Dec 2013	3,574	5.4 (0.2, 26.2)	65,578	406,148	2.8 (2.0, 3.9)	11,344	5.8 (0.8, 30.8)	-	N	-	-
Upper East Region, Dec 2013	1,472	25.9 (11.6, 50.2)	5,694	139,516	2.7 (1.8, 4.1)	3,787	1.5 (0.7, 3.4)	-	N	-	-
Upper West Region, Dec 2013	916	7.7 (0.2, 36.2)	11,896	94,036	3.1 (2.0, 4.7)	2,932	4.1 (0.5, 24.0)	-	N	-	-
Liberia	10						5.5 (4.1, 7.5)				
Bomi County, May 2016	623	24.8 (15.6, 37.0)*	2,512	15,906	2.4 (1.3, 4.6)	388	6.5 (3.0, 14.2)	1	N	N	N
Bong County, May 2016	1,593	24.8 (15.6, 37.0)*	6,423	63,055	1.7 (0.8, 3.6)	1,079	6.0 (2.6, 14.9)	1	N	N	N
Gbarpolu County, May 2016	376	24.8 (15.6, 37.0)*	1,516	25,703	1.9 (1.0, 3.8)	495	3.1 (1.4, 7.1)	1	N	N	N
Grand Cape Mount County, May 2016	591	24.8 (15.6, 37.0)*	2,383	15,767	2.9 (1.6, 5.1)	454	5.2 (2.4, 11.0)	1	N	N	N
Lofa County, May 2016	2,073	24.8 (15.6, 37.0)*	8,359	41,918	1.4 (0.5, 3.6)	570	14.7 (4.7, 43.6)	1	N	N	N
Margibi County, May 2016	827	24.8 (15.6, 37.0)*	3,335	23,684	3.4 (1.6, 7.0)	801	4.2 (1.8, 9.7)	1	N	N	Y
Maryland County, May 2016	942	24.8 (15.6, 37.0)*	3,798	52,349	1.5 (0.6, 4.0)	796	4.8 (1.7, 14.2)	1	N	N	N
River Gee County, May 2016	316	24.8 (15.6, 37.0)*	1,274	19,396	0.4 (0.1, 3.1)	86	14.8 (1.7, 62.0)	1	N	N	N
Rivercess County, May 2016	459	24.8 (15.6, 37.0)*	1,851	12,629	1.6 (0.6, 4.2)	200	9.3 (3.0, 24.9)	1	N	N	N
Sinoe County, May 2016	375	24.8 (15.6, 37.0)*	1,512	24,027	0.8 (0.2, 3.2)	197	7.7 (1.7, 28.2)	1	N	N	N
Mali	32						4.6 (3.7, 5.6)				
Région de Bamako, Aug 2017	4,073	34.9 (25.7, 44.4)*	11,658	380,700	2.7 (1.9, 3.7)	10,240	1.1 (0.7, 1.8)	1	Y	-	-
Région de Gao, Aug 2017	9,441	34.9 (25.7, 44.4)*	27,023	114,300	3.2 (2.1, 5.0)	3,675	7.4 (4.4, 12.7)	2	Y	-	-
Région de Kayes, Aug 2017	14,646	34.9 (25.7, 44.4)*	41,921	419,400	4.0 (3.1, 5.2)	16,937	2.5 (1.8, 3.7)	2	Y	-	-
Région de Kidal, Aug 2017	481	34.9 (25.7, 44.4)*	1,377	14,400	1.2 (0.5, 3.2)	173	8.0 (2.9, 22.2)	1	Y	-	-
Région de Koulikoro, Aug 2017	16,415	34.9 (25.7, 44.4)*	46,984	509,400	2.8 (2.0, 3.9)	14,104	3.3 (2.2, 5.2)	2	Y	-	-
Région de Mopti, Aug 2017	24,582	38.5 (28.5, 48.4)	63,913	428,400	1.8 (1.2, 2.8)	7,809	8.2 (4.9, 13.3)	2	Y	-	-
Région de Ségou, Aug 2017	20,701	34.9 (25.7, 44.4)*	59,252	492,300	3.6 (2.6, 5.1)	17,741	3.3 (2.1, 5.2)	1	Y	-	-
Région de Sikasso, Aug 2017	13,524	34.9 (25.7, 44.4)*	38,709	556,200	2.6 (2.0, 3.4)	14,387	2.7 (1.9, 4.0)	1	Y	-	-
Région de Tombouctou, Aug 2017	11,391	34.9 (25.7, 44.4)*	32,604	142,200	2.9 (1.8, 4.6)	4,115	7.9 (4.7, 13.1)	2	Y	-	-

ESTIMATING THE NUMBER OF CHILDREN IN NEED OF TREATMENT FOR WASTING

Location & date	Severe Wasting treatment admissions (n)	Coverage estimate (%)	Estimated incident cases (n)	Population (n)	Prevalence (%)	Prevalent cases (n)	K-value for severe wasting estimates	IPC phase at the time of data collection	Prevalence estimate taken during lean season? (Y/N)	Moderate Wasting treatment programme present?	Acute emergency (Y/N)
Région de Bamako, Jul 2016	5,183	34.9 (25.7, 44.4)*	14,835	372,600	2.5 (1.7, 3.7)	9,400	1.6 (1.0, 2.5)	1	Y	-	-
Région de Gao, Jul 2016	9,700	34.9 (25.7, 44.4)*	27,764	111,600	3.8 (2.7, 5.2)	4,189	6.6 (4.4, 10.0)	2	Y	-	-
Région de Kayes, Jul 2016	16,642	34.9 (25.7, 44.4)*	47,634	410,400	1.6 (0.9, 2.7)	6,413	7.4 (3.9, 14.1)	1	Y	-	-
Région de Koulikoro, Jul 2016	18,832	34.9 (25.7, 44.4)*	53,902	498,600	2.1 (1.4, 3.1)	10,448	5.2 (3.2, 8.5)	1	Y	-	-
Région de Mopti, Jul 2016	24,517	34.9 (25.7, 44.4)*	70,174	419,400	2.8 (1.9, 4.2)	11,792	6.0 (3.6, 10.2)	1	Y	-	-
Région de Ségou, Jul 2016	21,820	34.9 (25.7, 44.4)*	62,454	481,500	5.0 (3.8, 6.6)	24,125	2.6 (1.8, 3.9)	1	Y	-	-
Région de Sikasso, Jul 2016	15,770	34.9 (25.7, 44.4)*	45,138	544,500	1.9 (1.2, 3.3)	10,617	4.3 (2.4, 7.5)	1	Y	-	-
Région de Tombouctou, Jul 2016	12,985	34.9 (25.7, 44.4)*	37,166	138,600	4.4 (3.2, 6.1)	6,156	6.0 (4.0, 9.2)	2	Y	-	-
Région de Bamako, May 2015	6,971	34.9 (25.7, 44.4)*	19,953	364,500	1.8 (1.0, 3.0)	6,482	3.1 (1.7, 6.0)	1	N	-	-
Région de Gao, May 2015	10,065	34.9 (25.7, 44.4)*	28,809	108,900	2.5 (1.8, 3.7)	2,776	10.4 (6.7, 16.9)	2	N	-	-
Région de Kayes, May 2015	19,222	32.0 (21.4, 42.9)	60,156	401,400	3.7 (2.7, 5.1)	14,983	4.0 (2.6, 6.6)	1	N	-	-
Région de Koulikoro, May 2015	26,685	44.4 (33.9, 55.0)	60,064	480,600	2.8 (1.9, 4.3)	13,643	4.4 (2.7, 7.3)	1	N	-	-
Région de Mopti, May 2015	27,779	34.9 (25.7, 44.4)*	79,511	409,500	2.0 (1.3, 3.2)	8,307	9.6 (5.8, 16.9)	1	N	-	-
Région de Ségou, May 2015	26,573	36.5 (25.9, 47.1)	72,846	470,700	1.6 (0.9, 2.8)	7,403	9.8 (5.1, 18.1)	1	N	-	-
Région de Sikasso, May 2015	20,326	34.9 (25.7, 44.4)*	58,178	531,900	3.6 (2.5, 5.0)	19,039	3.1 (2.0, 4.7)	1	N	-	-
Région de Tombouctou, May 2015	14,892	34.9 (25.7, 44.4)*	42,625	135,900	4.4 (3.3, 5.9)	5,983	7.1 (4.8, 10.8)	2	N	-	-
Région de Bamako, Jul 2014	5,721	34.9 (25.7, 44.4)*	16,375	357,300	3.8 (2.6, 5.6)	13,637	1.2 (0.7, 1.9)	1	Y	-	-
Région de Kayes, Jul 2014	19,143	34.9 (25.0, 45.1)	54,900	393,300	2.1 (1.2, 3.5)	8,168	6.7 (3.6, 12.5)	2	Y	-	-
Région de Koulikoro, Jul 2014	24,696	29.3 (22.7, 36.0)	84,203	477,900	3.2 (2.1, 4.8)	15,193	5.5 (3.4, 8.9)	2	Y	-	-
Région de Mopti, Jul 2014	19,035	29.5 (21.6, 37.7)	64,423	401,400	3.0 (2.6, 3.4)	11,970	5.4 (4.1, 7.7)	2	Y	-	-
Région de Ségou, Jul 2014	24,811	41.9 (31.5, 52.3)	59,196	460,800	3.1 (2.1, 4.6)	14,303	4.1 (2.6, 6.6)	1	Y	-	-
Région de Sikasso, Jul 2014	20,133	28.1 (20.2, 36.0)	71,701	521,100	2.6 (2.3, 3.1)	13,673	5.2 (3.9, 7.6)	1	Y	-	-
Région de Tombouctou, Jul 2014	9,157	18.8 (13.1, 22.8)	48,707	133,200	3.4 (2.2, 5.2)	4,477	10.9 (6.8, 19.9)	2	Y	-	-
Mauritania	2						5.0 (0.9, 26.4)				
Gorgol Wilaya, Jun 2012	2,892	32.1 (19.9, 46.3)	9,009	59,914	1.2 (0.6, 2.6)	744	12.1 (5.3, 30.5)	-	Y	-	-
Hodh el Gharbi Wilaya, Jun 2012	911	33.1 (20.8, 48.0)*	2,748	51,548	2.4 (1.5, 3.9)	1,231	2.2 (1.2, 4.4)	-	Y	-	-
Niger	13						7.0 (5.5, 9.0)				
Agadez Region, Aug 2016	6,951	39.5 (30.3, 48.6)*	17,601	83,808	2.5 (1.7, 3.7)	2,090	8.4 (5.2, 13.5)	2	Y	Y	-
Dosso Region, Aug 2016	28,166	39.5 (30.3, 48.6)*	71,319	422,546	1.4 (0.7, 2.8)	6,050	11.8 (5.8, 25.0)	1	Y	Y	-

ESTIMATING THE NUMBER OF CHILDREN IN NEED OF TREATMENT FOR WASTING

Location & date	Severe Wasting treatment admissions (n)	Coverage estimate (%)	Estimated incident cases (n)	Population (n)	Prevalence (%)	Prevalent cases (n)	K-value for severe wasting estimates	IPC phase at the time of data collection	Prevalence estimate taken during lean season? (Y/N)	Moderate Wasting treatment programme present?	Acute emergency (Y/N)
Maradi Region, Aug 2016	108,988	39.5 (30.3, 48.6)*	275,967	761,168	4.3 (2.8, 6.5)	32,469	8.5 (5.3, 13.6)	1	Y	Y	-
Tillabery Region, Aug 2016	38,738	39.5 (30.3, 48.6)*	98,088	570,674	2.9 (1.7, 4.7)	16,266	6.0 (3.5, 10.3)	1	Y	Y	-
Zinder Region, Aug 2016	110,378	39.5 (30.3, 48.6)*	279,487	776,598	4.5 (2.9, 7.0)	35,135	8.0 (4.6, 13.0)	1	Y	Y	-
Agadez Region, Aug 2015	7,602	43.9 (35.2, 52.6)	17,335	82,411	2.9 (1.5, 5.6)	2,405	7.2 (3.6, 13.6)	1	Y	Y	-
Diffa Region, Aug 2015	16,577	39.5 (30.3, 48.6)*	41,974	107,355	3.8 (2.3, 6.1)	4,050	10.4 (6.3, 18.4)	3	Y	Y	-
Dosso Region, Aug 2015	31,211	39.5 (30.3, 48.6)*	79,029	90,830	6.1 (4.1, 8.9)	5,520	14.3 (9.2, 22.4)	1	Y	Y	-
Maradi Region, Aug 2015	97,523	39.5 (30.3, 48.6)*	246,937	737,693	7.0 (5.2, 9.5)	51,953	4.8 (3.3, 7.0)	1	Y	Y	-
Niamey Region, Aug 2015	7,136	39.5 (30.3, 48.6)*	18,069	168,128	4.3 (2.5, 7.3)	7,203	2.5 (1.4, 4.7)	1	Y	Y	-
Tahoua Region, Aug 2015	69,397	39.5 (30.3, 48.6)*	175,719	661,278	5.0 (3.1, 7.9)	32,816	5.4 (3.3, 9.4)	2	Y	Y	-
Tillabery Region, Aug 2015	36,820	56.3 (45.7, 66.9)	65,391	556,099	2.8 (1.6, 4.7)	15,405	4.2 (2.5, 7.2)	2	Y	Y	-
Zinder Region, Aug 2015	119,326	34.6 (25.2, 41.2)	344,584	752,585	5.1 (3.4, 7.7)	38,623	8.9 (5.9, 15.2)	2	Y	Y	-
Nigeria	47						2.3 (2.0, 2.7)				
Adamawa, Jul 2015	19,301	35.3 (26.3, 44.3)*	54,658	629,514	2.2 (1.0, 4.8)	13,636	4.0 (1.7, 10.0)	3	Y	-	-
Bauchi, Jul 2015	17,665	35.3 (26.3, 44.3)*	50,025	1,046,370	3.7 (2.5, 5.5)	38,874	1.3 (0.8, 2.2)	1	Y	-	-
Borno, Jul 2015	42,514	35.3 (26.3, 44.3)*	120,395	893,265	4.4 (3.0, 6.5)	39,264	3.1 (1.9, 4.9)	4	Y	-	-
Gombe, Jul 2015	11,407	35.3 (26.3, 44.3)*	32,303	517,124	4.4 (2.9, 6.7)	22,781	1.4 (0.9, 2.3)	1	Y	-	-
Jigawa, Jul 2015	70,175	35.3 (26.3, 44.3)*	198,728	986,637	4.1 (2.8, 5.9)	40,507	4.9 (3.2, 7.6)	1	Y	-	-
Kano, Jul 2015	57,195	35.3 (26.3, 44.3)*	161,970	2,042,267	4.3 (2.9, 6.4)	87,635	1.8 (1.2, 3.0)	1	Y	-	-
Katsina, Jul 2015	76,776	35.3 (26.3, 44.3)*	217,421	1,356,385	4.9 (3.4, 7.0)	66,557	3.3 (2.1, 5.3)	1	Y	-	-
Kebbi, Jul 2015	30,259	35.3 (26.3, 44.3)*	85,690	738,204	3.3 (2.2, 4.9)	24,471	3.5 (2.2, 5.9)	1	Y	-	-
Sokoto, Jul 2015	25,799	27.1 (19.5, 34.7)	95,208	840,495	5.3 (3.8, 7.3)	44,775	2.1 (1.4, 3.4)	1	Y	-	-
Yobe, Jul 2015	52,394	35.3 (26.3, 44.3)*	148,374	492,827	3.4 (2.3, 5.2)	16,994	8.7 (5.6, 14.5)	4	Y	-	-
Zamfara, Jul 2015	33,601	35.3 (26.3, 44.3)*	95,154	744,246	3.3 (2.2, 5.1)	24,848	3.8 (2.4, 6.1)	1	Y	-	-
Adamawa, Mar 2014	6,488	27.6 (18.3, 36.8)	23,549	614,220	2.0 (1.0, 3.9)	12,014	2.0 (0.9, 4.4)	2	N	-	-
Borno, Mar 2014	15,504	35.3 (26.3, 44.3)*	43,906	871,563	2.7 (1.6, 4.4)	23,429	1.9 (1.1, 3.4)	3	N	-	-
Gombe, Mar 2014	6,681	27.3 (17.4, 37.2)	24,476	504,561	3.4 (2.3, 5.1)	17,263	1.4 (0.8, 2.5)	1	N	-	-
Jigawa, Mar 2014	51,616	48.5 (41.1, 55.8)	106,519	962,667	5.3 (3.9, 7.3)	51,451	2.1 (1.5, 3.0)	1	N	-	-
Kano, Mar 2014	39,205	28.0 (17.9, 38.1)	140,045	1,992,650	5.8 (4.3, 7.9)	116,337	1.2 (0.8, 2.1)	1	N	-	-
Katsina, Mar 2014	78,856	32.4 (22.1, 42.6)	243,480	1,323,432	4.3 (3.0, 6.1)	56,574	4.3 (2.7, 7.5)	1	N	-	-
Kebbi, Mar 2014	26,431	30.5 (20.6, 40.3)	86,784	720,269	5.9 (4.4, 8.1)	42,833	2.0 (1.4, 3.5)	1	N	-	-

ESTIMATING THE NUMBER OF CHILDREN IN NEED OF TREATMENT FOR WASTING

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Sokoto, Mar 2014	20,493	48.3 (41.7, 55.0)	42,402	820,075	3.4 (2.3, 5.1)	27,928	1.5 (1.0, 2.4)	1	N	-	-
Yobe, Mar 2014	34,262	39.7 (30.7, 48.7)	86,370	480,854	3.5 (2.3, 5.2)	16,774	5.1 (3.3, 8.1)	3	N	-	-
Zamfara, Mar 2014	30,349	39.6 (30.7, 48.5)	76,718	726,164	2.8 (1.8, 4.4)	20,685	3.7 (2.3, 6.3)	1	N	-	-
Katsina, Sep 2013	54,934	53.4 (43.9, 62.9)	102,876	1,291,279	6.7 (5.2, 8.6)	86,601	1.2 (0.9, 1.6)	1	Y	-	-
Jigawa, Aug 2013	45,412	40.3 (30.7, 49.8)	112,796	939,279	5.4 (4.0, 7.2)	50,431	2.2 (1.5, 3.4)	1	Y	-	-
Kano, Aug 2013	36,931	35.3 (26.3, 44.3)*	104,584	1,944,238	4.8 (3.4, 6.6)	93,104	1.1 (0.7, 1.7)	1	Y	-	-
Sokoto, Aug 2013	11,776	35.3 (26.3, 44.3)*	33,348	800,151	3.6 (2.5, 5.1)	28,501	1.2 (0.7, 1.9)	1	Y	-	-
Yobe, Aug 2013	23,230	34.8 (23.7, 45.8)	66,786	469,172	2.4 (1.5, 3.7)	11,033	6.1 (3.5, 10.9)	3	Y	-	-
Zamfara, Aug 2013	23,907	35.3 (26.3, 44.3)*	67,702	708,522	4.0 (2.9, 5.7)	28,636	2.4 (1.6, 3.7)	1	Y	-	-
Adamawa, Jul 2013	5,218	35.3 (26.3, 44.3)*	14,777	599,298	1.4 (0.7, 2.5)	8,266	1.8 (0.9, 3.7)	1	Y	-	-
Gombe, Jul 2013	6,346	35.3 (26.3, 44.3)*	17,971	492,303	1.1 (0.6, 2.1)	5,511	3.3 (1.6, 7.1)	1	Y	-	-
Kebbi, Jul 2013	17,777	35.3 (26.3, 44.3)*	50,342	702,770	2.7 (1.8, 4.2)	19,071	2.6 (1.6, 4.6)	1	Y	-	-
Jigawa, Sep 2012	40,642	35.3 (26.3, 44.3)*	115,094	916,459	5.0 (3.7, 6.7)	45,767	2.5 (1.7, 3.8)	2	Y	-	-
Kano, Sep 2012	24,087	35.3 (26.3, 44.3)*	68,212	1,897,003	3.2 (2.2, 4.6)	60,843	1.1 (0.7, 1.7)	1	Y	-	-
Katsina, Sep 2012	49,373	35.3 (26.3, 44.3)*	139,819	1,259,907	3.5 (2.5, 4.8)	43,719	3.2 (2.1, 5.0)	1	Y	-	-
Kebbi, Sep 2012	19,333	35.3 (26.3, 44.3)*	54,749	685,697	5.6 (4.3, 7.4)	38,735	1.4 (1.0, 2.1)	1	Y	-	-
Yobe, Sep 2012	23,978	35.3 (26.3, 44.3)*	67,903	457,773	3.8 (2.6, 5.4)	17,331	3.9 (2.5, 6.3)	2	Y	-	-
Zamfara, Sep 2012	17,005	35.3 (26.3, 44.3)*	48,156	691,309	5.6 (4.3, 7.4)	39,013	1.2 (0.9, 1.9)	1	Y	-	-
Katsina, Mar 2012	49,373	35.3 (26.3, 44.3)*	139,819	1,259,907	1.6 (1.0, 2.7)	20,321	6.9 (4.0, 12.4)	1	N	-	-
Sokoto, Mar 2012	18,621	35.3 (26.3, 44.3)*	52,733	780,712	4.4 (3.1, 6.2)	34,525	1.5 (1.0, 2.4)	1	N	-	-
Yobe, Mar 2012	23,978	35.3 (26.3, 44.3)*	67,903	457,773	4.1 (2.9, 5.8)	18,877	3.6 (2.4, 5.6)	1	N	-	-
Kano, Feb 2012	24,087	35.3 (26.3, 44.3)*	68,212	1,897,003	2.8 (1.9, 4.2)	53,437	1.3 (0.8, 2.1)	1	N	-	-
Kebbi, Feb 2012	19,333	35.3 (26.3, 44.3)*	54,749	685,697	3.9 (2.7, 5.6)	26,524	2.1 (1.3, 3.3)	1	N	-	-
Zamfara, Feb 2012	17,005	35.3 (26.3, 44.3)*	48,156	691,309	4.0 (2.8, 5.6)	27,388	1.8 (1.2, 2.8)	1	N	-	-
Katsina, Aug 2011	45,048	35.3 (26.3, 44.3)*	127,571	1,229,298	4.5 (3.3, 6.1)	55,152	2.3 (1.6, 3.5)	1	Y	-	-
Kebbi, Aug 2011	14,687	35.3 (26.3, 44.3)*	41,592	669,038	3.4 (2.4, 4.8)	22,994	1.8 (1.2, 2.8)	1	Y	-	-
Sokoto, Aug 2011	16,261	35.3 (26.3, 44.3)*	46,049	761,744	4.2 (3.0, 5.9)	32,173	1.4 (0.9, 2.2)	1	Y	-	-
Yobe, Aug 2011	11,528	33.0 (23.8, 42.2)	34,944	446,652	1.9 (1.2, 3.0)	8,535	4.1 (2.4, 7.0)	1	Y	-	-
Jigawa, Jul 2011	23,375	35.3 (26.3, 44.3)*	66,195	894,194	5.1 (3.7, 6.9)	45,391	1.5 (1.0, 2.2)	1	Y	-	-

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Eastern and Southern Africa	87						3.7 (3.2, 4.3)				
Angola	1						6.4 (3.1, 14.5)				
Cubal & Ganda Municipalities, Mar 2017	3,401	26.2 (13.9, 42.0)	12,981	93,283	2.2 (1.3, 3.6)	2,028	6.4 (3.1, 14.5)	-	Y	Y	N
Ethiopia †	26						6.1 (4.9, 7.6)				
Abaala Woreda, Dec 2016	269	48.8 (37.4, 60.4)*	551	7,143	0.7 (0.3, 2.0)	53	10.4 (4.0, 28.6)	2	Y	Y	N
Adaar Woreda, Oct 2016	627	48.8 (37.4, 60.4)*	1,284	8,899	1.3 (0.6, 2.9)	115	11.2 (4.9, 25.8)	2	N	Y	Y
Bona Zuria Woreda, Oct 2016	501	48.8 (37.4, 60.4)*	1,026	20,969	1.1 (0.5, 2.3)	223	4.6 (2.0, 10.8)	1	N	N	N
Meiso Woreda, Aug 2016	1,118	50.9 (37.8, 63.7)	2,196	25,538	2.2 (1.4, 3.6)	564	3.9 (2.3, 7.0)	3	Y	Y	Y
Midhega Tola Woreda, Aug 2016	951	48.8 (37.4, 60.4)*	1,948	14,463	2.0 (1.1, 3.6)	290	6.7 (3.7, 12.9)	3	Y	Y	Y
Miyo Woreda, Aug 2016	251	48.8 (37.4, 60.4)*	514	9,849	0.4 (0.1, 1.7)	43	12.0 (2.7, 43.4)	2	Y	N	Y
Siraro Woreda, Aug 2016	2,703	48.8 (37.4, 60.4)*	5,537	27,186	1.7 (0.9, 3.1)	450	12.3 (6.7, 24.1)	3	Y	Y	Y
Raya Azebo Woreda, Jul 2016	283	48.8 (37.4, 60.4)*	580	20,349	1.3 (0.6, 2.5)	258	2.2 (1.1, 4.4)	3	Y	Y	N
Saesi Tsaeda Emba Woreda, Jul 2016	307	48.8 (37.4, 60.4)*	629	20,997	1.1 (0.5, 2.4)	229	2.7 (1.2, 6.5)	3	Y	Y	Y
Tankua Abergele Woreda, Jul 2016	398	48.8 (37.4, 60.4)*	815	13,709	1.2 (0.6, 2.3)	168	4.9 (2.5, 9.9)	3	Y	Y	Y
Abaala Woreda, May 2015	478	48.8 (37.4, 60.4)*	979	7,072	2.6 (1.6, 4.2)	182	5.4 (3.2, 9.6)	2	Y	Y	N
Adaar Woreda, May 2015	1,422	48.8 (37.4, 60.4)*	2,913	8,811	2.5 (1.5, 4.3)	223	13.1 (7.3, 24.0)	2	Y	Y	Y
Bona Zuria Woreda, May 2015	1,109	48.8 (37.4, 60.4)*	2,272	20,761	3.3 (2.1, 5.2)	687	3.3 (2.0, 5.7)	1	Y	N	N
Dessie Zuria Woreda, May 2015	525	48.8 (37.4, 60.4)*	1,075	21,903	1.4 (0.6, 3.1)	306	3.5 (1.5, 8.1)	2	N	Y	N
Gubalafto Woreda, May 2015	409	48.8 (37.4, 60.4)*	838	16,821	1.4 (0.6, 3.3)	232	3.6 (1.4, 8.9)	2	N	Y	Y
Halaba Woreda, May 2015	2,258	48.8 (37.4, 60.4)*	4,625	42,880	1.1 (0.5, 2.3)	481	9.6 (4.5, 22.0)	3	Y	Y	Y
Konso Woreda, May 2015	1,592	48.8 (37.4, 60.4)*	3,261	36,967	1.4 (0.7, 2.8)	521	6.3 (2.9, 12.8)	1	Y	Y	N
Meiso Woreda, May 2015	2,878	48.8 (37.4, 60.4)*	5,895	25,284	2.1 (1.2, 3.5)	526	11.2 (6.5, 20.3)	3	N	Y	Y
Midhega Tola Woreda, May 2015	1,428	48.8 (37.4, 60.4)*	2,925	14,319	2.4 (1.4, 4.2)	349	8.4 (4.7, 15.7)	3	N	Y	Y
Raya Azebo Woreda, May 2015	275	48.8 (37.4, 60.4)*	563	20,147	0.7 (0.2, 1.8)	134	4.2 (1.5, 10.3)	2	N	Y	N
Saesi Tsaeda Emba Woreda, May 2015	405	48.8 (37.4, 60.4)*	830	20,788	1.5 (0.8, 2.9)	318	2.6 (1.3, 5.4)	2	N	Y	Y
Sekota Zuria Woreda, May 2015	1,962	42.2 (30.0, 54.3)	4,652	19,466	3.0 (1.9, 4.7)	590	7.9 (4.7, 13.6)	3	N	Y	Y

ESTIMATING THE NUMBER OF CHILDREN IN NEED OF TREATMENT FOR WASTING

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Siraro Woreda, May 2015	3,516	48.8 (37.4, 60.4)*	7,202	26,916	1.8 (1.0, 3.2)	481	15.0 (7.8, 28.5)	3	N	Y	Y
Tankua Abergele Woreda, May 2015	329	48.8 (37.4, 60.4)*	674	13,573	1.0 (0.5, 2.0)	142	4.7 (2.3, 9.9)	3	N	Y	Y
Wenago Woreda, May 2015	1,613	48.8 (37.4, 60.4)*	3,304	24,371	1.1 (0.5, 2.4)	262	12.6 (5.7, 32.5)	1	Y	N	N
Bulehora Woreda, Mar 2014	1,422	48.8 (37.4, 60.4)*	2,913	55,796	1.0 (0.4, 2.3)	531	5.5 (2.2, 13.7)	2	N	N	N
Kenya	24						2.7 (2.1, 3.7)	-	-	-	
Isiolo County, Feb 2018	678	48.8 (36.9, 60.8)*	1,389	23,735	3.5 (2.3, 5.2)	820	1.7 (1.0, 2.9)	3	Y	-	-
Kajiado County, Jan 2018	2,117	48.8 (36.9, 60.8)*	4,337	130,965	3.1 (2.0, 4.8)	4,067	1.1 (0.7, 1.8)	3	-	-	-
Samburu County, Jun 2017	905	48.8 (36.9, 60.8)*	1,854	41,190	4.4 (2.9, 6.5)	1,794	1.0 (0.6, 1.6)	3	-	-	-
Turkana Central Subcounty, Jun 2017	2,695	48.8 (36.9, 60.8)*	5,521	15,308	9.5 (7.4, 11.9)	1,447	3.8 (2.7, 5.5)	3	-	-	-
Turkana North Subcounty, Jun 2017	834	48.8 (36.9, 60.8)*	1,708	15,479	9.6 (7.6, 12.2)	1,489	1.1 (0.8, 1.6)	3	-	-	-
Turkana South Subcounty, Jun 2017	1,876	48.8 (36.9, 60.8)*	3,843	9,984	13.2 (10.6, 16.2)	1,314	2.9 (2.2, 4.2)	3	-	-	-
Turkana West Subcounty, Jun 2017	7,846	48.8 (36.9, 60.8)*	16,072	30,960	8.6 (6.5, 11.4)	2,669	6.0 (4.2, 8.9)	3	-	-	-
Garissa County, May 2017	2,607	48.8 (36.9, 60.8)*	5,340	169,560	1.9 (1.1, 3.1)	3,215	1.7 (1.0, 2.9)	3	N	-	-
Laisamis Subcounty, May 2017	762	54.6 (41.9, 67.3)	1,395	8,798	7.1 (5.0, 10.0)	624	2.2 (1.5, 3.5)	3	N	-	-
Moyale Subcounty, May 2017	841	48.8 (36.9, 60.8)*	1,723	11,594	1.4 (0.6, 3.3)	162	10.6 (4.2, 26.9)	3	N	-	-
North Horr Subcounty, May 2017	785	48.8 (36.9, 60.8)*	1,608	9,501	6.9 (4.8, 9.8)	654	2.5 (1.6, 3.9)	3	-	-	-
Saku Subcounty, May 2017	350	53.0 (38.4, 67.6)	661	8,069	1.6 (0.6, 4.1)	126	5.2 (1.9, 13.7)	3	-	-	-
Turkana Central Subcounty, Feb 2017	2,921	48.8 (36.9, 60.8)*	5,984	15,308	8.2 (6.5, 10.3)	1,260	4.7 (3.4, 6.7)	3	-	-	-
Turkana North Subcounty, Feb 2017	1,253	48.8 (36.9, 60.8)*	2,567	15,479	9.4 (7.4, 11.7)	1,450	1.8 (1.3, 2.5)	3	-	-	-
Turkana South Subcounty, Feb 2017	2,018	48.8 (36.9, 60.8)*	4,134	9,984	7.2 (5.1, 10.1)	720	5.7 (3.8, 8.8)	3	-	-	-
Turkana West Subcounty, Feb 2017	8,347	48.8 (36.9, 60.8)*	17,098	30,960	4.7 (3.1, 6.9)	1,447	11.8 (7.5, 20.0)	3	-	-	-
West Pokot County, Mar 2013	2,620	21.7 (12.7, 30.7)	12,074	86,110	1.3 (0.6, 2.7)	1,129	10.7 (4.9, 27.0)	2	Y	-	-

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Isiolo County, May 2012	332	42.0 (26.0, 58.0)	790	19,571	3.7 (2.5, 5.4)	726	1.1 (0.7, 2.0)	2	N	-	N
West Pokot District, May 2012	1,038	33.5 (23.0, 44.0)	3,098	83,117	2.5 (1.4, 4.6)	2,088	1.5 (0.8, 3.1)	2	N	-	-
Makueni District, Mar 2012	405	48.8 (36.9, 60.8)*	830	120,806	0.4 (0.1, 1.6)	482	1.7 (0.4, 6.4)	2	N	-	N
Kitui District, Apr 2011	1,022	48.8 (36.9, 60.8)*	2,094	135,601	1.2 (0.6, 2.3)	1,602	1.3 (0.6, 3.0)	2	N	-	Y
Makueni District, Apr 2011	1,152	48.8 (36.9, 60.8)*	2,360	118,438	0.3 (0.1, 1.4)	413	5.7 (1.4, 19.0)	2	N	-	Y
Mwingi District, Apr 2011	1,010	48.8 (36.9, 60.8)*	2,069	48,607	1.7 (0.9, 3.1)	816	2.5 (1.4, 4.8)	2	-	-	Y
Mwingi District, Oct 2009	1,484	48.8 (36.9, 60.8)*	3,040	45,439	2.0 (1.2, 3.3)	899	3.4 (1.9, 5.7)	4	-	-	N
Madagascar	7						7.9 (6.4, 9.6)				
District de Betioky, May 2017	1,259	28.7 (20.9, 37.6)	4,387	46,504	1.9 (1.2, 3.0)	888	4.9 (2.8, 8.3)	3	N	Y	Y
District d'Ambovombe, Apr 2017	3,929	28.7 (20.9, 37.6)*	13,690	53,334	2.4 (1.6, 3.7)	1,297	10.6 (6.5, 17.7)	2	N	Y	Y
District d'Ampanihy, Apr 2017	3,270	28.7 (20.9, 37.6)*	11,394	62,357	1.7 (1.1, 2.5)	1,057	10.8 (6.5, 17.9)	3	N	Y	Y
District de Bekily, Apr 2017	1,794	28.7 (20.9, 37.6)*	6,251	33,059	2.3 (1.5, 3.6)	767	8.1 (4.8, 13.8)	2	N	Y	Y
District d'Amboasary, Mar 2017	2,930	28.7 (20.9, 37.6)*	10,209	52,665	2.5 (1.6, 3.9)	1,335	7.6 (4.6, 13.0)	3	Y	Y	Y
District de Taolagnaro, Mar 2017	1,608	28.7 (20.9, 37.6)*	5,603	62,434	1.2 (0.6, 2.5)	735	7.6 (3.4, 15.8)	3	Y	N	Y
District de Beloha, Feb 2017	1,665	28.7 (20.9, 37.6)*	5,801	21,991	3.8 (2.6, 5.3)	825	7.0 (4.6, 11.0)	3	Y	Y	Y
Somalia	1						2.5 (1.7, 3.7)				
Beledweyne Rural, Jul 2016	1,435	55.6 (45.2, 65.9)*	2,582	17,084	6.1 (4.4, 8.5)	1,047	2.5 (1.7, 3.7)	-	N	-	-
South Sudan	5						2.6 (1.8, 3.8)				
Ikotos County, Nov 2017	1,441	40.8 (29.6, 53.5)*	3,532	27,393	3.5 (2.3, 5.3)	967	3.7 (2.1, 6.0)	3	N	Y	Y
Tonj North County, Jul 2017	2,060	40.8 (29.6, 53.5)*	5,049	45,558	5.4 (4.1, 7.1)	2,447	2.1 (1.4, 3.2)	3	Y	Y	Y
Ikotos County, Jan 2017	1,580	40.8 (29.6, 53.5)*	3,873	27,393	9.1 (6.8, 12.1)	2,496	1.6 (1.0, 2.4)	3	N	Y	Y
Gogrial East County, Jun 2016	2,622	40.8 (29.6, 53.5)*	6,426	27,403	8.0 (5.9, 10.8)	2,204	2.9 (2.0, 4.6)	3	Y	Y	Y
Tonj North County, Jun 2016	1,616	40.8 (29.6, 53.5)*	3,961	43,948	2.0 (1.2, 3.3)	861	4.6 (2.5, 8.3)	2	Y	Y	Y
Uganda	23						2.8 (2.4, 3.2)				
Abim district, Karamoja Region, Jul 2018	564	34.4 (29.6, 40.7)*	1,640	19,468	2.8 (1.9, 4.2)	541	3.0 (1.9, 4.8)	1	Y	Y	N
Kaabong district, Karamoja Region, Jul 2018	1,153	34.4 (29.6, 40.7)*	3,352	30,272	5.9 (4.3, 8.0)	1,786	1.9 (1.3, 2.6)	2	Y	Y	N
Kotido district, Karamoja Region, Jul 2018	2,056	34.4 (29.6, 40.7)*	5,977	32,647	4.1 (2.8, 5.8)	1,324	4.5 (3.1, 6.6)	2	Y	Y	N

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Moroto district, Karamoja Region, Jul 2018	750	34.4 (29.6, 40.7)*	2,180	18,651	3.3 (2.2, 5.0)	620	3.5 (2.2, 5.7)	2	Y	Y	N
Nakapiripirit district, Karamoja Region, Jul 2018	827	34.4 (29.6, 40.7)*	2,404	28,254	3.5 (2.0, 5.9)	979	2.5 (1.4, 4.3)	2	Y	Y	N
Napak district, Karamoja Region, Jul 2018	644	34.4 (29.6, 40.7)*	1,872	25,646	3.8 (2.9, 4.8)	965	1.9 (1.4, 2.6)	2	Y	Y	N
Abim district, Karamoja Region, Dec 2017	564	34.4 (29.6, 40.7)*	1,640	18,874	2.4 (1.5, 3.6)	445	3.7 (2.3, 5.7)	2	N	Y	Y
Kaabong district, Karamoja Region, Dec 2017	1,153	34.4 (29.6, 40.7)*	3,352	29,348	4.0 (2.9, 5.5)	1,163	2.9 (2.0, 4.0)	2	N	Y	Y
Kotido district, Karamoja Region, Dec 2017	2,056	34.4 (29.6, 40.7)*	5,977	31,651	4.8 (3.6, 6.5)	1,533	3.9 (2.8, 5.4)	1	N	Y	Y
Moroto district, Karamoja Region, Dec 2017	750	34.4 (29.6, 40.7)*	2,180	18,082	5.1 (3.7, 7.0)	920	2.4 (1.7, 3.4)	2	N	Y	Y
Nakapiripirit district, Karamoja Region, Dec 2017	827	34.4 (29.6, 40.7)*	2,404	27,392	4.4 (3.1, 6.2)	1,208	2.0 (1.3, 2.9)	2	N	Y	Y
Napak district, Karamoja Region, Dec 2017	644	34.4 (29.6, 40.7)*	1,872	24,863	2.8 (1.9, 4.2)	698	2.7 (1.7, 4.2)	2	N	Y	Y
Abim district, Karamoja Region, Dec 2016	565	34.4 (29.6, 40.7)*	1,642	18,299	4.3 (2.6, 6.9)	781	2.1 (1.3, 3.5)	2	N	Y	Y
Kaabong district, Karamoja Region, Dec 2016	1,982	34.4 (29.6, 40.7)*	5,762	28,453	5.1 (3.2, 8.0)	1,459	3.9 (2.5, 6.1)	2	N	Y	Y
Nakapiripirit district, Karamoja Region, Dec 2016	1,161	34.4 (29.6, 40.7)*	3,375	26,556	4.7 (3.5, 6.3)	1,245	2.7 (1.9, 3.8)	2	N	Y	Y
Napak district, Karamoja Region, Dec 2016	853	34.4 (29.6, 40.7)*	2,480	24,105	7.2 (5.3, 9.8)	1,745	1.4 (1.0, 2.0)	2	N	Y	Y
Abim district, Karamoja Region, May 2016	466	34.4 (29.6, 40.7)*	1,355	18,299	1.5 (0.9, 2.6)	277	4.9 (2.7, 8.4)	2	Y	Y	Y
Amudat district, Karamoja Region, May 2016	339	34.4 (29.6, 40.7)*	985	17,926	3.3 (2.0, 5.3)	591	1.7 (1.0, 2.7)	2	Y	Y	Y
Kaabong district, Karamoja Region, May 2016	2,078	34.4 (29.6, 40.7)*	6,041	28,453	4.0 (2.9, 5.6)	1,140	5.3 (3.6, 7.6)	3	Y	Y	Y

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Kotido district, Karamoja Region, May 2016	1,868	34.4 (29.6, 40.7)*	5,430	30,685	6.0 (4.6, 7.9)	1,851	2.9 (2.1, 3.9)	2	Y	Y	Y
Moroto district, Karamoja Region, May 2016	965	34.4 (29.6, 40.7)*	2,805	17,530	5.0 (3.5, 7.2)	881	3.2 (2.1, 4.9)	3	Y	Y	Y
Nakapiripirit district, Karamoja Region, May 2016	1,287	34.4 (29.6, 40.7)*	3,741	26,556	4.7 (3.1, 6.9)	1,240	3.0 (1.9, 4.6)	2	Y	Y	Y
Napak district, Karamoja Region, May 2016	908	34.4 (29.6, 40.7)*	2,640	24,105	4.7 (3.3, 6.7)	1,133	2.3 (1.5, 3.5)	3	Y	Y	Y
Middle East and North Africa	34						2.9 (2.4, 3.5)				
Yemen	34						2.9 (2.4, 3.5)				
Abyan Governorate, Jan 2018	7,263	61.6 (51.5, 71.7)*	11,791	66,530	1.8 (1.2, 3.0)	1,229	9.6 (5.9, 15.1)	4	-	-	-
Lahj Governorate, Jul 2017	10,124	61.6 (51.5, 71.7)*	16,435	131,137	4.1 (3.0, 5.5)	5,316	3.1 (2.2, 4.4)	4	-	-	-
Ibb Governorate, Mar 2017	20,011	61.6 (51.5, 71.7)*	32,485	403,488	1.5 (0.9, 2.4)	5,934	5.5 (3.2, 9.5)	3	-	-	-
Shabwa Governorate, Jan 2017	5,558	61.6 (51.5, 71.7)*	9,023	99,590	1.3 (0.8, 2.1)	1,289	7.0 (4.1, 12.1)	4	-	-	-
Abyan Governorate, Dec 2016	6,838	61.6 (51.5, 71.7)*	11,101	64,192	3.1 (1.8, 5.1)	1,971	5.6 (3.3, 9.7)	4	-	-	-
Al-Baidha Governorate, Dec 2016	4,760	61.6 (51.5, 71.7)*	7,727	112,107	2.4 (1.3, 4.3)	2,693	2.9 (1.6, 4.9)	4	-	-	-
Al-Dale'e Governorate, Dec 2016	8,934	61.6 (51.5, 71.7)*	14,503	110,766	3.3 (2.0, 5.4)	3,692	3.9 (2.3, 6.5)	4	-	-	-
Al-Jawf Governorate, Dec 2016	3,249	61.6 (51.5, 71.7)*	5,274	99,266	4.6 (2.9, 7.1)	4,534	1.2 (0.7, 1.9)	3	-	-	-
Al-Mahweet Governorate, Dec 2016	6,655	61.6 (51.5, 71.7)*	10,804	96,332	4.5 (3.0, 6.7)	4,308	2.5 (1.6, 4.3)	3	-	-	-
Amran Governorate, Dec 2016	12,992	61.6 (51.5, 71.7)*	21,091	171,098	1.9 (1.0, 3.6)	3,303	6.4 (3.2, 12.1)	3	-	-	-
Dhamar Governorate, Dec 2016	19,937	61.6 (51.5, 71.7)*	32,365	345,036	4.8 (3.2, 7.0)	16,496	2.0 (1.3, 3.0)	3	-	-	-
Hajja Governorate, Dec 2016	21,355	61.6 (51.5, 71.7)*	34,667	323,414	3.5 (2.2, 5.5)	11,304	3.1 (1.9, 5.0)	4	-	-	-
Houdeidah Governorate, Dec 2016	70,047	61.6 (51.5, 71.7)*	113,713	520,335	6.8 (4.8, 9.5)	35,237	3.2 (2.2, 4.9)	4	-	-	-
Ibb Governorate, Dec 2016	17,981	61.6 (51.5, 71.7)*	29,190	396,334	2.6 (1.4, 4.6)	10,282	2.8 (1.6, 5.1)	3	-	-	-
Lahj Governorate, Dec 2016	10,611	61.6 (51.5, 71.7)*	17,226	128,812	4.2 (2.7, 6.6)	5,430	3.2 (2.0, 5.2)	4	-	-	-
Mareb Governorate, Dec 2016	2,440	61.6 (51.5, 71.7)*	3,961	47,007	1.4 (0.7, 3.0)	678	5.8 (2.7, 12.3)	3	-	-	-
Rayma Governorate, Dec 2016	3,468	61.6 (51.5, 71.7)*	5,630	90,121	3.2 (1.9, 5.3)	2,867	2.0 (1.1, 3.5)	3	-	-	-
Sa'ada Governorate, May 2016	18,937	61.6 (51.5, 71.7)*	30,742	178,153	5.0 (3.9, 6.4)	8,968	3.4 (2.6, 4.7)	4	-	-	-
Sana'a Governorate, May 2016	10,843	61.6 (51.5, 71.7)*	17,602	236,959	4.4 (3.3, 5.7)	10,311	1.7 (1.2, 2.4)	3	-	-	-

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Taiz Governorate, May 2016	21,180	61.6 (51.5, 71.7)*	34,383	480,057	5.2 (4.3, 6.4)	25,063	1.4 (1.1, 1.8)	4	-	-	-
Al-Baidha Governorate, Oct 2015	6,504	61.6 (51.5, 71.7)*	10,558	110,120	1.0 (0.4, 2.1)	1,064	9.9 (4.4, 25.5)	4	-	-	-
Lahj Governorate, Oct 2015	6,746	61.6 (51.5, 71.7)*	10,951	126,528	3.6 (2.6, 5.0)	4,580	2.4 (1.7, 3.5)	4	-	-	-
Hajja Governorate, Sep 2015	28,958	61.6 (51.5, 71.7)*	47,010	317,680	4.3 (3.3, 5.6)	13,686	3.4 (2.5, 4.7)	4	-	-	-
Aden Governorate, Aug 2015	5,370	61.6 (51.5, 71.7)*	8,718	142,967	3.2 (2.4, 4.2)	4,505	1.9 (1.4, 2.7)	4	-	-	-
Houdeidah Governorate, Aug 2015	62,005	61.6 (51.5, 71.7)*	100,657	511,109	11.2 (9.0, 13.9)	57,143	1.8 (1.3, 2.4)	4	-	-	-
Lahj Governorate, Jul 2014	5,267	61.6 (51.5, 71.7)*	8,550	124,285	1.9 (1.3, 2.7)	2,302	3.7 (2.4, 5.7)	4	-	-	-
Sa'ada Governorate, Jun 2014	9,956	61.6 (51.5, 71.7)*	16,162	171,892	5.4 (4.4, 6.8)	9,360	1.7 (1.3, 2.3)	4	-	-	-
Hajja Governorate, May 2014	21,421	61.6 (51.5, 71.7)*	34,774	312,047	3.0 (2.2, 4.0)	9,370	3.7 (2.7, 5.3)	4	-	-	-
Taiz Governorate, May 2014	17,522	61.6 (51.5, 71.7)*	28,445	463,186	3.1 (2.5, 3.9)	14,429	2.0 (1.5, 2.6)	4	-	-	-
Houdeidah Governorate, Mar 2014	42,361	61.6 (51.5, 71.7)	68,768	502,048	4.7 (3.7, 5.9)	23,383	2.9 (2.2, 3.9)	3	-	-	-
Amran Governorate, Sep 2013	5,163	61.6 (51.5, 71.7)*	8,381	162,157	2.5 (1.9, 3.4)	4,080	2.1 (1.5, 3.0)	3	-	-	-
Al-Mahweet Governorate, Jun 2013	3,038	61.6 (51.5, 71.7)*	4,932	91,299	3.4 (2.6, 4.4)	3,109	1.6 (1.2, 2.2)	3	-	-	-
Dhamar Governorate, Mar 2013	6,352	61.6 (51.5, 71.7)*	10,312	327,007	2.5 (1.9, 3.3)	8,270	1.2 (0.9, 1.7)	-	-	-	-
Abyan Governorate, Jan 2013	10,805	61.6 (51.5, 71.7)*	17,541	60,838	3.2 (2.5, 4.1)	1,932	9.1 (6.6, 12.2)	4	-	-	-
South Asia	12						3.0 (2.1, 4.4)				
Afghanistan	12						3.0 (2.1, 4.4)				
Farah province, Feb 2017	2,257	42.6 (32.6, 52.6)*	5,295	83,447	2.7 (1.8, 4.2)	2,283	2.3 (1.4, 3.8)	-	N	-	-
Ghor province, Aug 2016	6,510	42.6 (32.6, 52.6)*	15,273	112,384	8.4 (6.5, 10.7)	9,447	1.6 (1.2, 2.3)	3	Y	-	-
Badghis province, Feb 2016	3,124	42.6 (32.6, 52.6)*	7,329	80,772	3.2 (2.2, 4.6)	2,591	2.8 (1.9, 4.3)	3	N	-	-
Ghazni province, Jan 2016	6,348	42.6 (32.6, 52.6)*	14,893	200,195	7.4 (5.9, 9.3)	14,814	1.0 (0.8, 1.4)	2	N	-	-
Laghman province; Mehterlam, Alingar, Alishing, Dawlat shah, Qarghayi districts, Dec 2015	6,312	42.6 (32.6, 52.6)*	14,808	70,676	5.7 (4.4, 7.4)	4,050	3.7 (2.6, 5.3)	-	N	-	-
Balkh province, Aug 2015	3,849	42.6 (32.6, 52.6)*	9,030	217,684	2.5 (1.4, 4.4)	5,476	1.6 (0.9, 3.1)	-	Y	-	-
Nuristan province; Parun, Noorgram, and Wama districts, Aug 2015	840	42.6 (32.6, 52.6)*	1,971	9,369	8.7 (6.9, 10.9)	815	2.4 (1.8, 3.5)	3	Y	-	-

ESTIMATING THE NUMBER OF CHILDREN IN NEED OF TREATMENT FOR WASTING

Location & date	Severe Wasting treatment admissions (n)	Coverage estimate (%)	Estimated incident cases (n)	Population (n)	Prevalence (%)	Prevalent cases (n)	K-value for severe wasting estimates	IPC phase at the time of data collection	Prevalence estimate taken during lean season? (Y/N)	Moderate Wasting treatment programme present?	Acute emergency (Y/N)
Khost province; Matoon (Khost), Tanya and Mandozi (Ismail khail) districts, Jun 2015	2,524	42.6 (32.6, 52.6)*	5,921	41,828	3.3 (2.4, 4.4)	1,366	4.3 (3.0, 6.4)	3	N	-	-
Paktia province; Chamkani, Ahmad abad, Said karam and Gardez districts, Jun 2015	3,960	53.3 (42.5, 64.4)	7,430	35,878	2.4 (1.6, 3.5)	855	8.7 (5.6, 13.2)	-	N	-	-
Paktika province; Mata Khan, Sharana, Yahakhail, Sar Hawza, and Khair kot (Zarghon Shair) districts, May 2015	3,477	42.6 (32.6, 52.6)*	8,157	29,254	3.5 (2.5, 4.9)	1,025	8.0 (5.3, 12.1)	3	N	-	-
Helmand province; Garamser district, Mar 2015	1,233	42.6 (32.6, 52.6)*	2,893	17,374	3.5 (2.9, 4.4)	616	4.7 (3.5, 6.5)	2	N	-	-
Nangarhar Province; Jalalabad, Surkhrod, Kama, Behsud, and Kuzkunar (Khewa) districts, Dec 2014	7,252	52.6 (41.6, 63.4)	13,787	97,873	4.9 (3.6, 6.6)	4,771	2.9 (2.1, 4.2)	2	N	-	-
East Asia and Pacific	3						5.9 (4.4, 8.1)				
Myanmar	3						5.9 (4.4, 8.1)				
Pauktaw IDP camp, Jul 2016	422	82.1 (71.6, 92.7)	514	3,946	1.5 (0.8, 2.8)	59	8.7 (4.5, 17.7)	-	Y	-	Y
Buthidung Township, Sep 2015	4,996	56.7 (48.3, 64.7)	8,811	48,505	2.9 (1.7, 4.7)	1,402	6.3 (3.7, 10.4)	-	Y	-	Y
Maungdaw Township, Sep 2015	6,581	29.7 (21.4, 37.9)	22,158	84,302	5.4 (3.9, 7.5)	4,567	4.9 (3.2, 7.8)	-	Y	-	Y